

OPEN ACCESS

EDITED BY

Kurniawan Saefullah,
Padjadjaran University, Indonesia

REVIEWED BY

Francesco Bianco,
University of Cassino, Italy
Moharana Choudhury,
Voice of Environment (VoE), India

*CORRESPONDENCE

Vishal Diwan
✉ vishal.diwan@ki.se;
✉ vishal.diwan@icmr.gov.in

[†]These authors have contributed equally
to this work

RECEIVED 01 October 2025

REVISED 24 January 2026

ACCEPTED 30 January 2026

PUBLISHED 09 March 2026

CITATION

Trushna T, Yadav M, Sahoo KC, Singh S,
Parashar V, Pandey AK, Sabde YD, Pati S,
Tiwari RR and Diwan V (2026)
Contextualising household waste
segregation in India's small cities: a
multi-stakeholder qualitative study.
Front. Sustain. Cities 8:1716184.
doi: 10.3389/frsc.2026.1716184

COPYRIGHT

© 2026 Trushna, Yadav, Sahoo, Singh,
Parashar, Pandey, Sabde, Pati, Tiwari and
Diwan. This is an open-access article
distributed under the terms of the
[Creative Commons Attribution License](#)
(CC BY). The use, distribution or
reproduction in other forums is
permitted, provided the original
author(s) and the copyright owner(s) are
credited and that the original publication
in this journal is cited, in accordance
with accepted academic practice. No
use, distribution or reproduction is
permitted which does not comply with
these terms.

Contextualising household waste segregation in India's small cities: a multi-stakeholder qualitative study

Tanwi Trushna^{1†}, Manju Yadav^{1†}, Krushna Chandra Sahoo^{2,3},
Surya Singh^{1,4}, Vivek Parashar⁵, Ashok Kumar Pandey⁶,
Yogesh D. Sabde^{1,7}, Sanghamitra Pati³, Rajnarayan R. Tiwari^{1,7}
and Vishal Diwan^{1,7,8*}

¹ICMR-National Institute for Research in Environmental Health, Bhopal, India, ²Department of Health Research, Ministry of Health & Family Welfare, Government of India, New Delhi, India, ³ICMR-Regional Medical Research Centre, Bhubaneswar, Odisha, India, ⁴Faculty of Biological Sciences, Academy of Scientific & Innovative Research (AcSIR), Ghaziabad, India, ⁵R.D. Gardi Medical College, Ujjain, India, ⁶ICMR-Regional Medical Research Centre, Gorakhpur, Uttar Pradesh, India, ⁷Faculty of Medical Sciences, Academy of Scientific & Innovative Research, Ghaziabad, India, ⁸Department of Global Public Health, Karolinska Institutet, Stockholm, Sweden

Introduction: Household waste segregation is a central pillar of sustainable urban waste management, yet its everyday implementation in small cities in a low- to middle-income country like India remains uneven despite clear policy mandates and widespread awareness. This qualitative study examined how waste segregation is organised, negotiated, and stabilised in practice across three small cities in the country: Datia (Madhya Pradesh), Deoria (Uttar Pradesh), and Balasore (Odisha), selected to reflect diverse socio-cultural and infrastructural contexts.

Methods: Using an interpretive approach inspired by phenomenological design, 24 focus group discussions and 12 in-depth interviews were conducted with community residents and sanitation service providers. Data were audio-recorded, transcribed, and thematically analysed in MAXQDA using a reflexive approach.

Results: Five interrelated themes emerged, capturing how segregation practices were shaped by the temporal rhythms of daily life, lived material infrastructures, social logics and informal value systems, frontline sanitation workers' mediation, and flows of institutional legitimacy. Rather than functioning as a stable household behaviour, segregation emerged as a situational accomplishment, continuously negotiated at the interface of household routines, collection practices, and governance arrangements. City-level contrasts revealed that clearer procedural design, women-led frontline engagement, and sustained mediation supported more stable practices in Balasore, while procedural ambiguity and uneven service signals in Datia and Deoria contributed to conditional and symbolic compliance.

Discussion: The findings demonstrated that persistent gaps between policy intent and everyday practice arise less from deficits in awareness and more from mis-alignments between service design and lived realities. The study underscores the limitations of standardised, metro-derived, technology-intensive waste management models and highlights the need for small-city policies that align service delivery with everyday routines, recognise informal waste practices, strengthen frontline mediation, and build institutional legitimacy through visible and credible waste flows.

KEYWORDS

service providers, waste management, waste segregation, waste workers, women workers

1 Introduction

Household waste management has become a major concern for developing countries due to fast urbanisation, economic activities, industrialisation and population growth (Sarker et al., 2024; Lenkiewicz, 2024). These factors have significantly altered the dynamics of waste generation and management. Limitations in the collection of waste and its transportation have caused the accumulation of waste, dumpsites are expanding, and Indian smart city initiatives are being affected (Hamdan et al., 2025). According to the claims of a report by the International Solid Waste Association (ISWA), global waste production is expected to expand to 3.8 billion tonnes by 2050, which was only 2.1 billion in 2023 (Lenkiewicz, 2024). India, being one of the most populous countries, produces 62 million tonnes of municipal solid waste and is expected to reach 436 million tonnes by 2050 (Dhanuskar et al., 2024). The increasing volume of waste generated puts a serious strain on the municipalities and local bodies, whose capacities and resources often remain insufficient to meet the growing demand for waste management (Biswas et al., 2021). This difference in resource requirements and infrastructure contributes to persistent difficulties in proper waste management, making it a sustainability challenge. Environmental challenges arising from water, soil and air pollution, linked to waste management, are of increasing concern to sustainable development. Rise of microplastics, an emerging contaminant, especially around the waste disposal sites in India, emphasises the need for improved waste collection and separation to prevent accumulation in landfills and leaching into the land and water (Singh et al., 2025).

The waste management sector in India is growing significantly to achieve sanitation and cleanliness goals, with actions that aim to strengthen improved waste management in the country (International Trade Administration, 2023). Even after the implementation of the Clean India Mission (hereafter referred to as Swachh Bharat Mission-SBM) (Municipal Solid Wastes, 2023), limitations in waste collection efficiency and inadequate waste segregation at the source lead to a huge amount of waste ending up in landfills in big cities (for example, the “X” and “Y” cities defined in India by the central government as having a population size of over fifty lakhs and between five and fifty lakhs, respectively and a higher cost of living) (Srivastava and Chakma, 2024; Kalyanasundaram et al., 2023; Census-of-India-2001, 2014; Kumar K. et al., 2024). Anaerobic degradation of the mixed waste in un-engineered dumpsites is a major source of greenhouse gases (GHGs) (Srivastava and Chakma, 2024), posing health risks to communities, contaminating the environment and increasing the cost of waste management infrastructure (Velijala, 2024; Mondal and Mandal, 2024). Heavy metals, construction and demolition waste, dumpsites, and inappropriate dumping of waste in water and on land are major causes of soil and water pollution and contribute to the creation of breeding sites for disease-causing vectors. Dumpsite leachates are also known to be significant contributors to heavy metals in soil and water in both developed and developing countries, affecting public health and the environment (Frazer-Williams and Sankoh, 2024; Shukla et al., 2025).

Significant differences exist in the waste management needs and challenges faced by Indian cities, particularly between large and small ones. For example, unlike larger cities, small Indian cities and towns, having a population of less than five lakhs, generate relatively less waste, often less than 100 tonnes per day (Census-of-India-2001, 2014). However, these small cities are also facing difficulties in

managing waste due to constraints related to resources, technical capacity, and locally specific institutional and social conditions (Ahluwalia and Patel, 2018). Thus, these cities potentially require context-appropriate technological and organisational approaches, while in larger cities, management of the vast amounts of generated solid waste is difficult and requires modernised mechanizations for waste treatment (Census-of-India-2001, 2014; Zhang et al., 2024). Furthermore, community participation in waste management/segregation varies significantly in smaller cities due to various factors, including regional, socio-demographic differences, amount and type of waste generation and embedded practices and understandings of waste (Leknoi et al., 2024). Therefore, the one-size-fits-all strategy may not adequately address the challenges of waste management in small cities and may require approaches tailored to each city's size and local contexts (Rai et al., 2024). However, there is limited published evidence of how waste segregation practices are enacted, reasoned, and sustained in everyday life in small Indian cities.

The current study was designed to examine how community members and service providers experience, reason, and enact waste management practices in small Indian cities, with particular attention to the everyday conditions under which waste segregation is sustained, negotiated, or disrupted. Rather than treating waste management as a purely technical or behavioural problem, the study adopted an interpretive perspective that views waste management as a socially embedded process shaped by routine household practices, material arrangements, institutional structures, and frontline service interactions. Waste management in small cities involves multiple actors, including households, sanitation workers, service providers, and local authorities, whose roles, responsibilities, and constraints intersect in practice rather than operating in isolation. Adopting a multi-stakeholder perspective, therefore, enabled a more nuanced understanding of how waste segregation is organised in everyday urban life, how responsibilities are interpreted and mediated across actors, and why particular forms of waste handling persist. By foregrounding these lived practices and perspectives, the study generates empirically grounded insights that can inform more context-sensitive approaches to waste management in small Indian cities.

2 Methods

2.1 Study setting and design

This study employed an exploratory qualitative design, informed by an interpretive perspective drawing on phenomenological principles, to understand how community members and waste-management service providers experience and make sense of household waste segregation in small Indian cities. Rather than applying a strict phenomenological protocol, our approach prioritised understanding shared lived experiences, perceptions, practical reasoning, and context-specific meanings associated with waste segregation practices across a heterogeneous group of stakeholders with diverse roles, contexts, and institutional positions. The analytical process followed a thematic analysis, coupled with ongoing reflexivity and iterative team discussions, enabling the systematic identification and refinement of patterns across diverse stakeholder groups and study sites, thereby representing waste segregation practices as they are enacted and negotiated in three geographically diverse small Indian cities.

Three cities namely Datia (Madhya Pradesh, Central India), Deoria (Uttar Pradesh, Northern India), and Balasore (Odisha, Eastern India) were purposively selected to incorporate the variations in local contexts and related stakeholder perspectives in three geographically disparate regions of the country, with significant differences in sociocultural practices (Figure 1). To fulfill the study objectives, we chose small (category “Z”) cities that were not included in the list of big or “X”/“Y” cities, which are defined by the Indian central government as having a population size of over five lakhs (i.e., 500,000) and higher cost of living. Another criterion for selection was that these cities were included in the list of cities where Swachh Survekshan is routinely conducted. Swachh Survekshan is an annual assessment and ranking of selected Indian cities that evaluates the effectiveness of their municipal solid waste management, conducted under the auspices of SBM. Table 1 provides an overview of these three cities.

Datia is a historical district of India and secured 93rd position in the Swachh Survekshan ranking in the year 2023 (Swachh Survekshan Ranking, 2024). The city generates 30.08 tonnes per day (TPD) of waste, with residents practising partial source segregation. According to the action plan of the city, it requires information, education and communication (IEC) material to achieve 100% source segregation (Datia Environment Plan, 2021). However, the city ranked 85th in 2022, indicating a need to improve waste management practices at the service and community levels. Deoria is a small city in the eastern part of Uttar Pradesh, bordering Bihar. The city secured 147th position in the 2023 Swachh Survekshan Ranking (Swachh Survekshan Ranking,

2024). The city has improved its position from 212 in 2020 to 152 in 2021. These shifts indicate changing assessed performance over time, providing a relevant context for examining how waste management practices are organised and experienced locally. Balasore is located in the extreme north of the Odisha state and occupied 99th position among the cleanest cities in India in the 2023 Swachh Survekshan Ranking (Swachh Survekshan Ranking, 2024). Agriculture is the primary occupation in the region, followed by fishing and tourism (Government of Odisha, n.d.). Waste collection in the city is the responsibility of the Urban Local Bodies (ULBs), but there is no information on whether 100% door-to-door collection of waste is carried out. Additionally, mini trucks provided for garbage collection do not have separate compartments for different waste types (Baleswar Action Plan, 2023).

2.2 Study participants

The study participants were selected using purposive sampling to ensure diverse representation across stakeholder types (community residents and waste management service providers) and geographical regions (Datia, Deoria, Balasore). Community members, including males and females living in slums and non-slums, were included because their everyday experiences and interpretations of waste management may differ from those of service-providing stakeholders. The other category of stakeholders was the frontline waste collection workers (waste collectors and waste collection vehicle drivers), supervisory and management personnel (sanitary inspectors, trenching

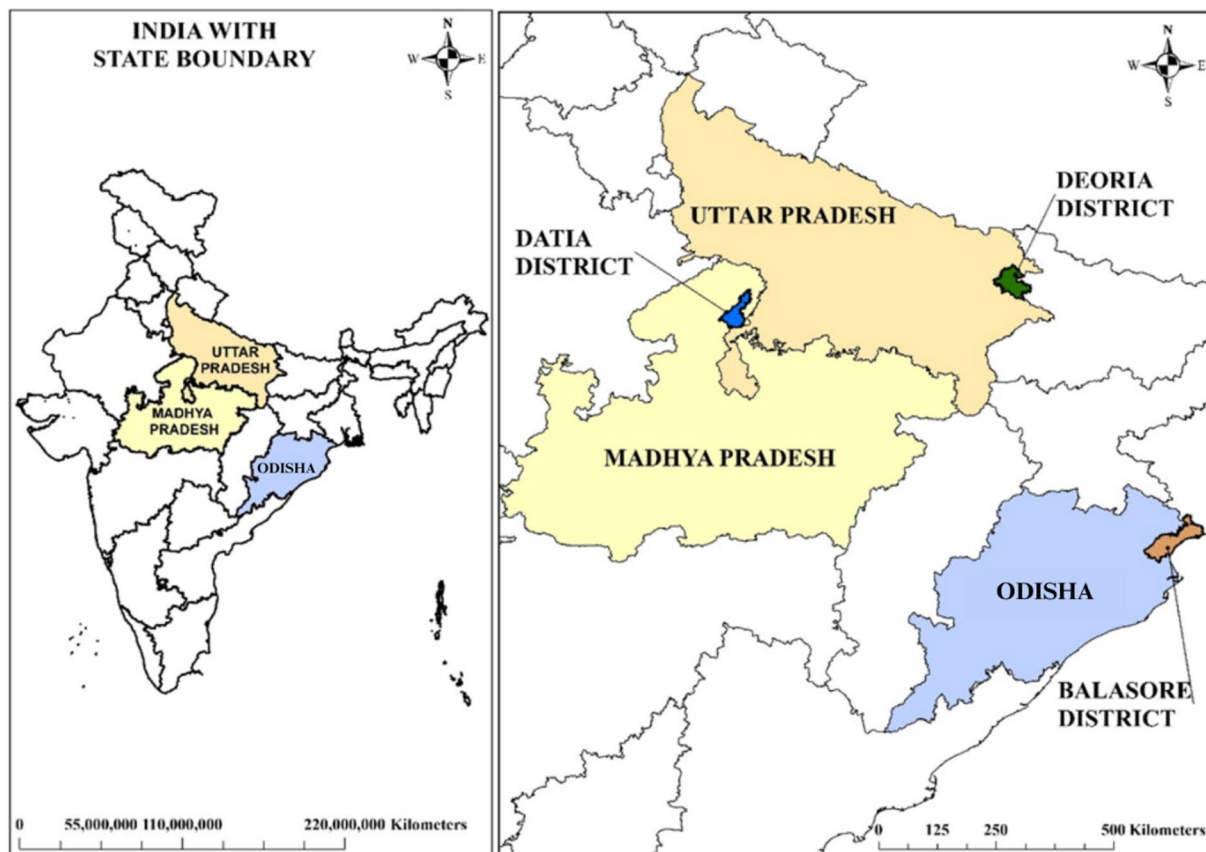


FIGURE 1
Diagrammatic representation of three sites selected for the purposive study.

TABLE 1 Population and waste generation data of study settings.

Characteristics of the city	Datia	Deoria	Balasore
Total population	181,982	316,803	253,293
Male population	96,191	1,64,325	1,29,321
Female population	85,791	1,52,478	1,23,972
Total waste generated*	30.08	96.38	75.74
Swachh Suvekshan ranking 2022	85	147	126
Swachh Suvekshan ranking 2023	45	No information	99

* Denotes the total waste generation in tonnes per day (TPD) reported by [Census of India-2011 \(2015\)](#).

supervisors, program managers, and waste-related service providers) and workers at waste management and recycling facilities (waste management volunteers and material recovery facility (MRF) workers) (Table 2). These stakeholders were selected because they were directly involved in, or closely engaged with, the everyday organisation, delivery, and mediation of waste management practices, and were therefore well positioned to reflect on how waste segregation is enacted in practice.

In Balasore city, we identified another set of stakeholders comprising Swachhata Sathi, Swachha karmis, and battery-operated vehicle (BOV) drivers, who were appointed to manage household waste in residential areas alongside or as alternatives to the conventional waste collectors and waste collection vehicle drivers. The Swachhata sathis were female sanitation workers assigned to oversee waste collection and awareness activities in a residential ward. Swachhakarmis were women who accompanied garbage vehicles, collected waste from houses, and segregated it into its categories if it had not been previously segregated. BOV drivers were women who operated battery-powered garbage vehicles after completing the municipality's proper driving and garbage collection training. These posts have been created under the Mission Shakti to empower women. In addition, some female volunteers were community residents with prior experience and familiarity of waste segregation at source, and their work was supervised by program managers and cleaning supervisors, who also ensured proper door-to-door waste collection and other related activities. We included many of these female workers as respondents during our qualitative data collection.

2.3 Data collection procedure

A total of 24 focus group discussions (FGDs) among community members and frontline waste collection workers, and 12 in-depth interviews (IDIs) among supervisory and management personnel were conducted. We conducted seven FGDs and seven IDIs in Datia, 12 FGDs and 3 IDIs in Deoria, and 5 FGDs and 2 IDIs in Balasore (Table 2). Among these FGDs and IDIs, respondents' ages were widely distributed. Approximately 62% were male, and most participants were educated to at least the primary level. Women respondents worked as ASHA workers, shopkeepers, tailors, or housewives, while men were commonly drivers, labourers, shopkeepers, small business owners, and farmers. These discussions enabled the development of a nuanced understanding of how waste management practices are experienced

TABLE 2 Demographic details of the participants from Balasore, Datia and Deoria.

City	Total FGD/ IDI	Age	Mean $\pm$ std dev	Gender (M/F)
Balasore	5 FGD	19–85	39.7 $\pm$ 11.4	9/31
	3 IDI	28–34	31 $\pm$ 4.2	2/1
Datia	7 FGD	18–72	40.12 $\pm$ 15.8	29/20
	7 IDI	28–35	31 $\pm$ 2.8	7/0
Deoria	12 FGD	21–71	37 $\pm$ 12	55/13
	3 IDI	37–43	40 $\pm$ 3	3/0
Participants				171

and interpreted, as the participants came from diverse demographic backgrounds. All FGDs and IDIs were facilitated by a multidisciplinary team of qualitative researchers, comprising both male and female investigators fluent in the local languages. The interviewers/facilitators had prior training and experience in qualitative data collection and analysis. Researchers came from diverse academic backgrounds, including medical sciences, clinical practice, and public health, with educational qualifications ranging from bachelor's to doctoral degrees. Their professional roles similarly spanned scientists affiliated with the Government of India's environmental health research institutions, public health professionals, and practising medical clinicians. This diversity in educational, professional, and sociodemographic profiles contributed to a richness of perspectives throughout the study design, data collection, and interpretation. While every effort was made to maintain neutrality during data collection, ongoing reflexivity was practised throughout to minimise any potential bias.

The discussions and interviews lasted usually for around 35–40 min. No interviews or focus group discussions were repeated with the same participants. Efforts were made to ensure that non-participants were not present during data collection, in order to maintain confidentiality and protect participant anonymity. Among the participants, the service provider participants were contacted face-to-face and/or telephonically through their workplace whereas the community residents were approached through community leaders. Only a few individuals declined participation, primarily due to scheduling conflicts or competing personal and work commitments. No participants dropped out once they had consented to take part in the study. After obtaining consent, one-on-one interviews and group discussions were scheduled at a time convenient to the participants and conducted in person at a location of their choosing, often at the nearest community hall or Anganwadi centre. Although no overt prior personal relationship was established between the researchers and participants before the commencement of the study, basic rapport building was done through multiple interactions before the interviews/discussions and participants were approached during fieldwork after obtaining institutional permissions. Participants were informed that the researchers were affiliated with medical college, public health and environmental research institutions and that the purpose of the study was to understand how household waste segregation is experienced and organised in everyday contexts.

The FGD and IDI guides were prepared using open-ended questions with probes (Table 3). The guides were translated into the regional language. These guides were pilot-tested, and the team was familiarised with them before data collection began. The FGDs and

TABLE 3 Focus group discussion (FGD) and In-depth interview (IDI) guide*.

SN	Questions	Probes
1	Describe your household waste management.	How do you currently separate your household waste? Please describe: • changes or adjustments in the current household waste segregation system over time; • measures to establish composting or biodegradable waste systems; • disposal of household hazardous waste, such as batteries and electronics; • local landfills that accept these waste types; • changes in waste segregation practices compared to the past.
2	What obstacles have you encountered in properly sorting and discarding household waste?	How have you responded to the obstacles to the proper sorting and discarding of household waste in everyday practice? How can you reduce the amount of waste your household generates to minimise the need for sorting and disposal? Have you considered using reusable bags, containers, and other eco-friendly options to reduce household waste?
3	What recommendations do you have for your community or local government to improve waste management and segregation in your city?	NA
4	How do you ensure your recyclables are properly separated and not contaminated with other waste types?	NA
5	How can you make the separation of household waste a fun and educational activity for children and other family members?	What is the role of males and females in household waste segregation? How can you encourage other household members to participate in waste segregation and make it a household-wide initiative?
6	What is the provision for household waste management in your area?	Public dustbins, Provision of separate dustbins for households, Garbage collecting vans, Collection time, Human resources, Private organisations' involvement (PPP mode), Community support, Involvement of community-based organisations, Role of informal service provided
7	Can you please explain the system of household waste collection and management in your locality?	How is household waste collected? How is it segregated? Distance of the local collection point from the household What is done with the collected segregated waste? Where is the waste taken after collecting it from the households? What are the time schedules for emptying and replacing public bins?
8	What is your reflection on current household segregation practices?	Perceived Strengths and Limitations, Reflections on periodic feedback from the general public, Reflections on how these practices are sustained or challenged in everyday service delivery, Suggestions for improving household waste segregation practices.

* The topic guides were used flexibly to facilitate open-ended discussion, with probes adapted during interviews and group discussions to follow participants' experiences, interpretations, and locally relevant practices.

IDIs were conducted until data saturation. All the FGDs and IDIs were audio recorded.

2.4 Data management and analysis

The audiotaped FGDs and IDIs were transcribed verbatim from the regional language to English by trained research assistants. Field notes from the interviews/discussions were also used, along with the audio recordings, for transcription. Owing to the wide variation in

the geographic areas where interviews/discussions were conducted, all transcripts could not be returned to participants for their opinions and feedback. Codes were formed using open coding. Transcripts were coded using MAXQDA 24 software by two researchers working simultaneously, and differences in interpretation were discussed with the rest of the team. The data were analysed through thematic analysis. Similar codes were categorised together, and multiple categories were merged to create themes from the collected data. The codes, categories, and themes were read by the authors TT, MY, VD, and

KCS to avoid errors and misinterpretation of the information. All the authors contributed to the data analysis to enhance the confirmability of the findings.

2.5 Ethical consideration

The study was conducted after obtaining approval from the Institutional Ethics Committee of ICMR-National Institute for Research in Environmental Health (NIREH), Bhopal. All participants were treated with respect concerning their values, rights and interests. No human experiments or changes in any clinical treatment were included in the proposed project. All data were analysed and reported at the aggregate level only; no individual participant data were released. No personal identifiers were used in the questionnaire except sex and age. Researchers emphasised before all assessments that data collection aimed to gather information to understand everyday waste management practices, not to impose sanctions or punishments. Participants were briefly informed

about the project, the participant information sheet was provided, and written consent was obtained before recording their views and insights on waste management practices in their respective cities.

3 Results

The five interrelated themes emerged, which are: (1) Temporal Rhythms and Everyday Synchronisation, (2) Lived Infrastructures and Material Arrangements, (3) Social Logics, Value Systems, Gendered Roles, and Informal Practices, (4) Sanitation Workers as Practice Mediators in Frontline Waste Management, and (5) Institutional Embeddings and Legitimacy Flows. Together, the first three themes explain how households organise, reason through, and stabilise waste segregation in everyday life, under temporal, material, and social constraints (Table 4). The latter two themes elucidate how these household practices are mediated, enabled, constrained, and reworked through

TABLE 4 Coding tree with themes, categories and codes dealing with household-centred organisation of waste segregation practices.

Themes	Categories	Codes
Temporal rhythms and everyday synchronisation	1.1 Negotiating waste handover within time-pressured household morning routines	- Juggling time pressure and priorities in morning routines - Gendered dressing-related hesitation at collection time
	1.2 Coordinating waste handover at uncertain and signal-dependent collection encounters	- Coping with uncertainty in waste collection timing - Waste handover during short or missed collection van halts - Waste handover depending on sound cues
Lived infrastructures and material arrangements	2.1 Household bin arrangements as lived infrastructure	- Defaulting to single-bin or limited multi-bin household arrangements - Household bin distribution, demand perception and (dis)use
	2.2 Living with heterogeneous waste collection vehicle compartmentalisation	- Coexisting waste collection vehicle compartmentalisation designs - Segregation Constrained By Vehicle Design
	2.3 Spatial constraints and access workarounds	- Restricted vehicle reach shaping where waste is handed over - Residential space limits on storing segregated waste - Community bins as spatial workarounds for storage and access constraints
	2.4 Procedural ambiguities and their downstream consequences	- Procedural ambiguity shaping everyday waste sorting - Downstream handling shaped by mis-sorted waste
Social logics, value systems, gendered roles, and informal practices	3.1 Segregation practices oriented toward expected everyday effects	- Health-protective expectations motivating segregation - Sensory and environmental expectations of segregation - Cleanliness, social respectability, and care-based motivations
	3.2 Moral and normative dimensions of waste management	- Social norms and informal neighbourhood moral enforcement - Internalised moral self-regulation of waste practices - Community-side conditional moral accountability (system-contingent) - Provider-side conditional institutional accountability (user-contingent) - Symbolic and performative compliance
	3.3 Informal waste practices operating beyond the municipal waste stream	- Wet waste diversion to livestock - Conditional adoption of composting practices - Selective segregation shaped by informal value logics
	3.4 Gendered and age-ordered organisation of household waste practices	- Routinisation of waste handover as women's work - Men's indirect oversight and instruction in waste practices - Age-differentiated participation in waste handover practices

service delivery arrangements, frontline labour, and broader governance structures shaping waste management in small cities (Table 5).

3.1 Temporal rhythms and everyday synchronisation

This theme describes how household waste segregation and handover are organised within the temporal rhythms of everyday life in small Indian cities. Rather than being a stand-alone or reflective

activity, waste segregation was embedded in tightly structured daily routines, particularly in the morning schedules shaped by domestic work, employment, schooling, and caregiving responsibilities.

3.1.1 Negotiating waste handover within time-pressured household morning routines

Participants described the waste handover as tightly constrained by early-morning routines, with collection timing overlapping with

TABLE 5 Coding tree with themes, categories and codes describing service provision, frontline mediation, and institutional embedding of waste segregation.

Sanitation workers as practice mediators in frontline waste management	4.1 Active doorstep mediation and real-time classification practices	- Mediating waste handover at the doorstep - Negotiating waste classification to prevent mixing at collection
	4.2 Organisational conditions shaping frontline mediation work	- Inconsistent training preparation for mediation work - Supervisory oversight shaping frontline work - Duty allocation and temporal pressures shaping mediation work
	4.3 Stigma, recognition, and financial empowerment shaping sanitation workers' participation	- Stigma shaping worker-household interactional boundaries - Women sanitation workers negotiating gendered stigma through public work visibility - Women sanitation workers' financial independence and dependency reversal as a work motivator
	4.4 Gendered structuring of frontline sanitation practices	- Gendered mediation of sensitive sanitation interactions - Gendered task allocation and skill acquisition in sanitation work
Institutional embeddings and legitimacy flows	5.1 Post-collection opacity and the erosion of institutional trust	Post-collection opacity in waste flows
	5.2 Negotiating service legitimacy through user fees and sanctioning practices	- User fees / challans as instruments of behavioural accountability - Contested legitimacy and social frictions in user fee implementation - Incremental normalisation of user fees through routine service delivery
	5.3 Everyday uptake of segregation-related conventional communication	- Heterogeneity of ongoing communication practices - incremental accumulation of waste-sorting Understandings through repeated communication - Limited translation of communicative messages into everyday sorting routines - Perceived limits of conventional communication in shaping everyday practice
	5.4 Adaptive reworking of communication practices to restore relevance	- Experimenting with alternative communication formats to re-engage communities - Reframing segregation through locally meaningful consequences and benefits - Seeding segregation norms through intergenerational educational pathways
	5.5 Hierarchical, socio-political, and administrative structuring of local waste governance	- Asymmetric hierarchical oversight and recognition in sanitation work - Informal political and social power mediating frontline waste governance - Upstream policy and administrative decisions structuring frontline practice
	5.6 Felt need and efforts to modify material and manpower resource infrastructure	- Material infrastructure conditions shaping segregation and processing practices - Vehicle availability and collection modalities influencing door-to-door operations - Dependence on manual labour under conditions of mixed waste flows
	5.7 Sustained effort, local adaptation, and interactional adjustments in community engagement	- Perceived effects of prolonged and repeated engagement with communities - Situated workarounds in response to spatial, temporal, and household constraints - Interactional norms and self-regulation in frontline service encounters - Symbolic reframing of waste spaces to shape public perception - Existing complaint resolution mechanisms and feedback pathways in everyday governance

activities such as bathing and eating. When collection vehicles arrived during these moments, households were often unable to respond immediately, leading to missed handovers as the vehicles moved on quickly. Fixed collection schedules, therefore, did not guarantee successful waste disposal, as bodily routines and household norms around cleanliness shaped whether waste could be handled at that time. Service providers also noted that reluctance to touch waste after bathing, among both residents and workers, further narrowed the effective collection window. Together, these accounts indicate that missed waste handovers were shaped by temporal misalignment and embodied practices rather than by intentional non-participation.

"When the garbage van comes, people are bathing or eating, due to which the van goes away, and people are not able to dump the garbage." (Deoria: Community Residents (Slum) Male FGD)

"There is a fixed timing of garbage trucks inside Datia. That leaves at 7 am in the morning. And after a person takes a bath, he says, 'we will not touch the garbage bucket', and some of the labourers also do not pick it up. (Datia: Service Provider (Frontline Sanitation Worker) FGD).

The morning waste handover was shaped by competing work and school schedules, which often conflicted with collection timing. Households with members leaving early for duty or school reported setting waste aside in advance or postponing disposal, while others were unable to wait for collection vehicles arriving later in the morning. These temporal conflicts were not framed as a refusal but as a practical prioritisation of employment and schooling obligations. From both community and service provider perspectives, work-related departures limited households' ability to respond to collection, and in some cases, collection constraints such as full vehicles further deferred handover.

"Yes, there is a lot of hurry in the morning. Children go to school. And whoever understands puts the garbage there." (Deoria: Community Residents (Non-Slum) Female FGD)

"We should also take care that those who go out to work set out the garbage [for collection beforehand]. If we say something to them, then they say [in response] that we have to go to do duty." (Deoria: Community Residents (Slum) Female FGD).

"And if a person's office time is 9:00 am and the garbage trucks come at 10:00 am, so he says that we will keep it outside in the evening, that's why he is not able to keep it." (Datia: Service Provider (Frontline Sanitation Worker) FGD)

"People [who go to] work, they say, the vehicle is full of garbage, we will go to the plant and [ask us to] come back later." (Balasore: Service Provider (Frontline Sanitation Worker) FGD)

Participants also noted that early-morning waste handover was shaped by the informality of dress at the time of collection. In the group discussion of supervisors from Deoria, it was explained that women engaged in household work were often in informal attire, making it difficult for them to step outside immediately when collection vehicles arrived. Participants anticipated that this might limit the ability to respond at short notice and contribute

to missed handovers, even when waste was otherwise ready for disposal.

"In the morning (at the time of waste collection), gents are sleeping, women are working in the house. So now she is wearing a maxi, etc... She has to come out in that and give over the waste." (Deoria: Service Provider (Supervisor) FGD)

3.1.2 Coordinating waste handover at uncertain and signal-dependent collection encounters

Building on the time pressures of the morning, participants described waste handover as unfolding within collection practices characterised by brief or variable vehicle halts. Service providers acknowledged that collection vehicles moved quickly through neighbourhoods, while drivers described managing limited stopping time to complete routes. Community members noted that vehicles sometimes moved past houses or altered direction even when residents were waiting with waste, shaping when and where handover could occur within already narrow morning windows. In this context, waste handover during morning collection was frequently conducted rapidly at the point of encounter, resulting in waste being transferred without careful separation in some instances.

"Whenever the garbage man comes! Have to give it quickly." (Deoria: Community Residents (Non-Slum) Female FGD)

"Some people are complaining that the vehicle is not going to their lane, only going to the main road, or the vehicle is going by at speed." (Balasore: Service Provider (Supervisor) FGD)

"I stand 2-5 minutes in front of each house." (Datia: Service Provider (Frontline Sanitation Worker) IDI)

"The vehicle comes, sir. But many times, it happens that we are standing with the garbage. And the driver turns the vehicle before our house and goes ahead." (Datia: Community Residents (Slum) Male FGD)

"When the van arrives, all the garbage gets mixed". (Deoria: Community Residents (Non-Slum) Male FGD)

Within these brief morning encounters, participants described sound cues as central to coordinating waste handover. For example, community members from Datia explained that horns, slogans, or recorded songs signalled the arrival of the collection vehicle and prompted households to immediately bring out their waste. Service providers similarly noted that sound cues were used to announce presence and trigger rapid handover without prolonged interruptions. When such cues were absent or unclear, residents reported difficulty anticipating arrival, increasing the likelihood that vehicles would pass without handover.

"How do you know it is the garbage vehicle... they give horns and slogans." (Datia: Community Residents (Slum) Female FGD)

"When the song plays, they [community] bring it [waste] automatically." (Datia: Service Provider (Frontline Sanitation Worker) IDI)

"Doesn't even sound the horn." (Datia: Community Residents Male (Slum) FGD)

However, reliance on auditory signals did not eliminate broader uncertainty around waste collection timing. Participants described uncertainty about waste collection timing as a recurring challenge households had to manage in their everyday routines. When collection vehicles did not arrive as expected, residents reported storing waste at home until the next day, treating delayed collection as something to be endured rather than resolved. Irregularities such as non-collection on specific days and inconsistent arrival times were perceived as routine features of service provision. From the community perspective, this unpredictability was linked to uneven enforcement of work schedules among sanitation staff, with collection times shifting across mornings and afternoons and work sometimes ending before completion.

"And if the driver does not come, then we have to keep the trash at home. Then it will come tomorrow." (Deoria: Community Residents (Non-Slum) Female FGD)

"Garbage car does not come on Sunday... should come sir". (Deoria: Community Residents (Non-Slum) Male FGD)

"The main problem here is that the [rules for] employees who are employed should be [strictly] imposed. If they have to work [properly], the whole city will be clean. They sometimes do cleaning, and sometimes not, they come sometimes in the morning to clean and sometimes at noon. If it is time to extract garbage, some employees leave the work before completion." (Datia: Community Residents (Slum) Male FGD).

3.2 Lived infrastructures and material arrangements

This theme examines how waste segregation practices are shaped through the everyday use, adaptation, and negotiation of household bins, collection vehicles, and urban spaces, highlighting how material arrangements are lived, negotiated, and worked around in small-city contexts.

3.2.1 Household bin arrangements as lived infrastructure

Participants described household waste storage as commonly organised around single-bin or limited multi-bin arrangements. Many community members reported using only one container for all waste, despite being aware of segregation guidelines, treating mixed storage as a practical default rather than a deliberate choice. Others pointed out that even when households attempted segregation, the design of collection vehicles with a single compartment reduced the perceived usefulness of maintaining separate bins at home.

"There are 25% of people who separate waste. The remaining 75% of the people mix the garbage and throw it away. Meaning they keep only one dustbin." (Datia: Community Residents (Slum) Male FGD)

"Here people do not keep the waste separately, they put all the waste in the same dustbin. Anyhow, only one compartment van comes. Now, whether you put wet or dry garbage in it, everything gets mixed." (Datia: Community Residents (Slum) Female FGD)

"There should also be two bins in the house to separate the garbage, but we cannot keep them. We keep the same box and put all the garbage in it." (Deoria: Community Residents (Non-Slum) Male FGD)

A smaller number of participants described keeping multiple containers for different waste types, often organising them by room or function within the household. Across accounts, household bin arrangements aligned with storage practices and expectations for how waste would ultimately be collected and handled.

"We keep three boxes, one in the kitchen, one for dry waste, and one for household waste." (Datia: Community Residents Female FGD, 230802_0258)

"Yes, we do it separately. There is a dustbin in the kitchen for vegetable waste. There is a separate dustbin for sweeping the house." (Datia: Community Residents Male FGD, 230801_0609)

Building on household storage practices, participants described how perceptions of bin availability and purpose shaped the use of distributed containers within homes. Service providers noted that even when multiple bins were distributed by municipal authorities, households often continued to use a single container, questioning the need for separate bins. Community members expressed mixed expectations regarding government provision, with some viewing bin supply as limited or impractical at scale, while others relied on existing household containers in place of formal dustbins. From a service-delivery perspective, this resulted in situations in which households symbolically displayed containers when asked but did not consistently maintain dedicated bins for different waste types.

"The Municipal Corporation also distributed three dustbins. They can separate the dry from the wet! But people believe that if there is a dustbin in our house, we will put all the garbage in it. Why do we buy/use two?" (Datia: Service Provider (Supervisor) IDI)

"We bought it [dustbin] ourselves for the house. It [the government's distribution of dustbins] would have been better. Can the government give everyone a dustbin in such a big area? For that, the government is sending the garbage truck in the morning and evening." (Balasore: Community Residents (Non-Slum) Female FGD)

"Yes, you go to ask for a dustbin from them, and they show two or three dustbins! So, there is a challenge: because they do not really have 2-3 dustbins! Someone waves to show a bath bucket! Someone brings something else, and we are facing a challenge with the team." (Datia: Service Provider (Supervisor) IDI)

3.2.2 Living with heterogeneous waste collection vehicle compartmentalisation

Extending these household-level interpretations to the point of collection, service providers described the coexistence of multiple waste collection vehicle designs within the same municipal systems. Across sites, community residents reported vehicles with varying degrees of compartmentalisation, ranging from fully segregated vehicles with dedicated compartments for multiple waste streams (mainly in Balasore and partially in Datia) to vehicles designed to carry mixed waste (mainly in Deoria and partially in Datia).

"Our vehicles have 2 compartments and another bag for sanitary [waste] purposes." (Balasore: Service Provider (Local Authority) IDI)

"We have two types of vehicles here! We have 35 vehicles in which there are four compartments. One for dry waste, one for wet waste, one for sanitary waste and one for hazardous waste... and apart from this, we have 15 vehicles which carry mixed waste as they do not have any compartment." (Datia: Service Provider (Supervisor) IDI)

"No, there is only one compartment; all kinds of garbage have to be dumped in it. Both wet and dry are inserted into it." (Deoria: Community Residents (Slum) Female FGD)

"You know that's why it's difficult for me to go to the vehicle to throw garbage, because if the vehicle is there [nearby], it will be easy for people from different houses to throw the garbage in it. In my case, the vehicle stopped near Jaspal's house, which is far from my house. So, it's difficult for me." (Balasore: Community Residents (Non-Slum) Male FGD)

Participants described residential space limitations as influencing how long and in what form waste could be stored within households. Service providers noted that shrinking dwelling sizes, particularly in flats or smaller homes, reduced the feasibility of retaining segregated waste over time. Community members similarly questioned where separated waste, especially dry or recyclable materials, could be kept within a limited living space.

"In today's time, the space area is decreasing, people are living in flats or in two-three rooms, there is no mentality of collecting much garbage. Earlier, they used to keep [waste] in the bigger houses they lived in; now they are throwing it away." (Deoria: Service Provider (Local Authority) IDI)

"If we separate the waste like packets [for recycling], then where should we keep it?" (Deoria: Community Residents (Non-Slum) Male FGD)

In response to limited storage space within homes, participants described community bins as a shared spatial workaround for waste disposal and access. Community members suggested centrally located bins to prevent indiscriminate dumping and reduce the need to store waste at home. However, accounts also highlighted that when community bins were infrequently emptied or poorly maintained, they quickly overflowed and attracted animals, undermining their intended function. Service providers questioned the fairness of penalising households in areas where such communal infrastructure was absent or inadequately spaced.

"It is better to put a big dustbin outside, have everyone put their garbage in it, and then, when the van comes, [they can] take that garbage out. This is the best solution." (Deoria: Community Residents (Slum) Males FGD)

"The dustbin gets filled in a single day, and the vehicle that used to come to pick up the garbage from that dustbin has been stopped by the municipality. It will get cleaned once a month. Animals also come out of it. You can now go to the road and see". (Datia: Community Residents (Slum) Males FGD)

3.2.3 Spatial constraints and access workarounds

Participants described how the physical reach of waste collection vehicles shaped where waste handover could occur. In several neighbourhoods, collection vehicles were reported to operate primarily along main roads, with limited access to inner lanes or smaller streets due to vehicle size or road conditions. As a result, households were often required to carry waste to designated collection points rather than have it picked up at their doorsteps. Community members noted that this distance made waste disposal more labour-intensive, particularly when vehicles had to stop far from their homes. Service providers similarly pointed to infrastructure limitations, such as incomplete or narrow roads, which influenced vehicle movement and, in turn, household access to collection.

"Vehicles ply from the main road only. Vehicles do not come to the by-lanes... Big van not able to go inside, small van could come. Big vehicles come only on the [main] road." (Datia: Community Residents (Slum) Female FGD)

"The area is big, just the road should be built. It has been sanctioned." (Deoria: Service Provider (Frontline Sanitation Worker) FGD).

"No, [waste collection vehicle] does not come in front of the house, comes on the main road... Then you go there to throw garbage..." (Datia: Community Residents (Slum) Female FGD)

3.2.4 Procedural ambiguities and their downstream consequences

Alongside spatial and infrastructural arrangements, participants described procedural ambiguities in everyday waste sorting that shaped how segregation was enacted at the household level. Community members reported uncertainty about how specific materials, such as plastic bags used to contain wet waste, should be classified at the point of disposal. Others noted that although waste categories were labelled on collection vehicles, these distinctions were not consistently explained, leaving households to rely on assumptions rather than clear guidance.

"The biggest problem here is the plastic bag. Here people use fewer plastic bags; and if, inside the dustbin, wet waste is kept in plastic bags, then throw the plastic bag directly [as wet waste]." (Datia: Community Residents (Slum) Male FGD)

"No, just the garbage van comes, its song plays.... No [explanation], because they [the volunteers of the municipality] especially come to do surveys.... It [hazardous waste category] is written on the garbage vehicle, but no one comes to explain." (Datia: Community Residents (Slum) Male FGD)

"Just as you told us which garbage to put in which colour compartment, the government should also tell the people, because people do not know which garbage to put in which colour box. When the van arrives, people put all the garbage together in the same compartment." (Deoria: Community Residents (Non-Slum) Male FGD).

Service providers described downstream waste-handling as shaped by the quality and composition of waste received from households. Mixed waste, including hazardous and sanitary materials, was reported to limit the feasibility of segregation at dumping or processing sites, with workers expressing hesitation in handling such waste. Participants noted that composting depended on receiving sufficiently clean wet waste, and when waste arrived wet and mixed, further separation became difficult or unsafe.

"We are not able to do segregation even there [in the dumping ground]. Because too much of the waste that we get is mixed, and it includes hazardous and sanitary waste, too! Then the employees also hesitate. If we get wet waste of quality, only then can we compost it. If quality wet waste is unavailable, it cannot be composted either. When a lot of garbage gets wet and mixed, it becomes difficult to separate it." (Datia: Service Provider (Supervisor) IDI)

3.3 Social logics, value systems, gendered roles, and informal practices

This theme reports on how waste segregation practices are shaped by expected effects, moral reasoning, social norms, informal value logics, and demographic roles, influencing who participates in segregation, under what conditions, and for what reasons in everyday small-city contexts.

3.3.1 Segregation practices oriented toward expected everyday effects

Participants described waste segregation as oriented toward expected everyday health and environmental effects. Both community members and service providers linked segregation and regular handover to reductions in visible pollution, foul odour, and the presence of flies and mosquitoes. In Balasore, service providers associated improved waste handling with fewer vector-related illnesses, while participants in Datia recalled earlier conditions of open dumping and blocked drains as directly connected to sickness in the community.

"Earlier, people were throwing garbage on the road, but now they are giving it to us. Because of that, the environment was getting polluted, it has reduced, and the flies and mosquitoes have also been killed." (Balasore: Service Provider (Supervisor) FGD)

"Then [before waste door-to-door collection] people were spreading filth in the streets, in the drains, on the roads. ... Drains are getting filled with garbage only. Because of this, people would also get sick." (Datia: Community Residents (Slum) Female FGD)

"Before that garbage was falling on the road and there was a lot of smell, but by taking that garbage away, it no longer smells." (Balasore: Service Provider (Supervisor) FGD)

Participants framed waste segregation as a practice tied to cleanliness, social respectability, and care for shared spaces, homes, and animals. Community members described strong aversion to visible litter, torn sanitary waste, foul odour, and the presence of animals scavenging in public spaces, recalling earlier conditions as unpleasant and socially undesirable. Segregation and tied disposal were associated with restoring walkability, reducing embarrassment when guests visited, and maintaining a sense of order within and outside the home. Participants also linked segregation to animal care, noting the harm caused when cows or dogs consumed mixed waste.

"People were throwing on the road and drain, the dog was tearing these pads and diapers and littering the road. We hated to walk on that road. Now, all these things are not happening. Now we tie it in polythene and throw it in the garbage truck." (Balasore: Community Residents (Non-Slum) Female FGD)

"Earlier, people used to throw away the vegetable peels in polythene, and the cows ate it all, including the polythene, harming them. ... but now they are instructed to open the polythene and to throw separately." (Balasore: Community Residents (Non-Slum) Female FGD)

"If we don't separate the wet and dry waste, then it will stink and cause diseases. Animals will be harmed. Yes, animals will also suffer." (Datia: Community Residents (Slum) Male FGD)

Interestingly, these care-based motivations also shaped how segregation was practically enacted, with some participants prioritising keeping household dustbins clean over strict adherence to segregation rules. In such cases, wet waste was intentionally placed in plastic bags to prevent odour and soiling of bins, reflecting a negotiation between cleanliness goals and prescribed sorting practices.

"Cannot put wet waste directly into the dustbin. If the peels and the remaining vegetables are put in the dustbin, it will stink. The dustbin will be dirty and smelly. The plastic bags went with wet waste, but the dustbin remained clean. It is just the biggest problem that the dustbin remains clean." (Datia: Community Residents (Slum) Male FGD)

3.3.2 Moral and normative dimensions of waste management

Participants described neighbourhood-level norms shaping acceptable waste practices through informal moral enforcement. Community members noted collective disapproval of burning waste due to smoke and perceived health harms, and reported intervening when individuals disposed of garbage outside designated collection systems.

"People used to bring garbage from their houses and throw it outside; we couldn't tell anyone. Now we are all throwing garbage in the garbage truck. And whoever is not throwing garbage properly in the garbage van, we are forcing him to throw garbage into the vehicle..." (Balasore: Community Residents (Non-Slum) Female FGD)

Additionally, participants described waste segregation as being increasingly guided by internalised moral self-regulation rather than solely by external enforcement. Community members noted that even when collection was delayed, households retained waste in their own containers instead of discarding it elsewhere, waiting to hand it over in the next collection cycle. This restraint was framed as a matter of conscience, shaped by cleaner surroundings and regular municipal services, which discouraged mixing or indiscriminate disposal.

"If on someday the garbage van does not come, the dustbin is left full. The next day, when the garbage truck comes, we dump it." (Datia: Community Residents (Slum) Female FGD)

"Now the sweeper is coming and cleaning the street. Earlier, people threw garbage in the drain; now, they [the sweeper] clean it. Now, the people's conscience is hindering them from doing so." (Balasore: Community Residents (Non-Slum) Female FGD)

Community members described responsibility for waste segregation as conditional on how collection systems operated in practice, often linking their own sorting efforts to vehicle design, reach, and staff prompts. Participants acknowledged segregation as desirable but questioned its relevance when collection vehicles lacked compartments or did not arrive at their doorstep, leading some to mix waste or to separate it only within the household. In these contexts, segregation was sometimes enacted performatively, with households demonstrating awareness or compliance during inspections or at the point of collection when explicitly instructed, rather than maintaining separation consistently.

"Here people do not keep the waste separately, they put all the waste in the same dustbin. Anyhow, only one compartment van comes. Now, whether you put wet or dry garbage in it, everything gets mixed." (Datia: Community Residents Female FGD, 230801_0514)

"When the garbage van comes, and [if] they say to throw the garbage separately, then we will throw it separately. [But] they don't say anything." (Datia: Community Residents (Slum) Female FGD)

Service providers interpreted these practices through a different lens, recognising gaps in service provision and communication while also attributing non-segregation to resident inertia or lack of sustained effort.

"They just need the excuse that we don't have to do it. Dustbin was given, some people [even] used homemade ones. No one kept the garbage segregated." (Deoria: Service Provider (Supervisor) IDI)

3.3.3 Informal waste practices operating beyond the municipal waste stream

Participants described wet waste in small cities as frequently diverted away from municipal collection through locally embedded practices. In Datia, both residents and service providers noted that vegetable peels and food leftovers were commonly fed to cows, resulting in lower volumes of wet waste entering the collection stream.

"The amount of wet waste is less! Because here, I have seen people throw away vegetables or peels. Throwing means feeding the animal or feeding it to the cow." (Datia: Service Provider (Supervisor) IDI)

"We only put dry waste! The wet garbage that is left for food, we feed the cow." (Datia: Community Residents (Slum) Female FGD)

In Balasore, participants more often described selective retention of wet waste for household composting and kitchen gardening, with non-compostable materials handed over to collection vehicles.

"We tell people that separate garbage to make compost." (Balasore: Service Provider (Frontline Sanitation Worker) FGD)

"I'm telling that we make a kitchen garden at our home where we use the wastewater that is used in our kitchen and make compost from vegetable peels, used flowers from our prayer room, and tea leftover, which is useful for the garden. We give waste like polythene to the garbage van as we cannot use it in compost." (Balasore: Community Residents (Non-slum) female FGD)

However, composting practices were uneven and conditional, shaped by prior knowledge, perceived usefulness, and local livelihood contexts. In Datia, participants noted that despite agrarian links, composting was less common due to limited instruction and the ready availability of dung manure, indicating that wet-waste segregation was guided by locally practical value rather than uniform municipal norms.

"It is a small city, people from rural areas are here, and more animal husbandry is here! No fertiliser required here. Dung manure is found here.... No, we have not seen that [using wet waste as manure]. And no one makes at home as it has not been taught how to make and neither has been told that you should give this wet waste to the plant." (Datia: Community Residents (Slum) Male FGD)

Participants described segregation as selectively applied to waste streams perceived to hold immediate or recoverable value. Both community members and service providers noted that dry waste, such as plastics or recyclables, was often retained, sold to scrap dealers, or

diverted for reuse, particularly when available in sufficient quantity. In contrast, materials viewed as low-value or inconvenient to store were more likely to be discarded with mixed waste.

"No, we sell all that in scrap, we do not mix it with the garbage, like a bottle of plastic". (Deoria: Community Residents (Slum) Female FGD)

"Whoever is going to take it! We sell it to them." (Datia: Service Provider (Supervisor) IDI)

"It [dry waste] was bought and taken away by a kabaddiwala [scrap dealer]. ...Yes, some people say that we will make manure, we will not give our garbage." (Balasore: Service Provider (Frontline Sanitation Worker) FGD)

[In response to the interviewer's remark: "Some municipalities do this on a large scale and also make a profit from it."] "Garbage is nothing gold, sir." (Deoria: Service Provider (Frontline Sanitation Worker) FGD)

3.3.4 Gendered and age-ordered organisation of household waste practices

Participants consistently described waste handover as routinised women's work within households, with women most often responsible for giving waste to collection vehicles. Men were described as less directly involved in handover, but sometimes exercised indirect oversight by instructing or encouraging women and other household members on how waste should be managed. This framing positioned men as decision-makers or advisors rather than routine actors in waste practices.

"Mostly women give the waste to the van". (Balasore: Community Residents (Non-Slum) Female FGD)

"Most women do it, but men also sometimes support and understand. How can the work be done well! So they understand and have an effect." (Datia: Community Residents (Slum) Male FGD)

"Gents are the main in our Indian scenario. If I tell the children/women in the house to take out the garbage like this, then they will be a little conscious about it." (Deoria: Service Provider (Supervisor) FGD)

Participants further noted that participation was also age-differentiated. Service providers noted difficulties in engaging older women with segregation instructions, while community members highlighted practical limits to children's involvement, such as school timing and physical ability to reach collection vehicles.

"And if we explain it to a 55-year-old woman, if we explain it to her, then she will not understand." (Deoria: Service Provider (Supervisor) FGD)

"We do. There are children in the house as well as men. The children cannot reach the vehicle." (Datia: Community Residents (Slum) Female FGD)

"Yes, children are intelligent, but most of the time, women dispose of garbage, and children go to school at that time, so we do not get [to meet] many children. We encourage it [children's involvement], but it all depends on their guardians. (Deoria: Service Provider (Frontline Sanitation Worker) FGD)

3.4 Sanitation workers as practice mediators in frontline waste management

This theme documents how sanitation workers actively mediate everyday waste practices in small cities through doorstep interaction, real-time classification, and negotiated engagement with households. It shows how organisational conditions, social stigma, public recognition, financial empowerment, and gendered task structuring shape workers' capacity to sustain these practices at the street level.

3.4.1 Active doorstep mediation and real-time classification practices

Service providers described actively mediating waste handover at the doorstep, combining signalling, explanation, and direct assistance to align household practices with collection procedures. Drivers and supervisors reported using sirens or verbal prompts to indicate segregation requirements, while also stepping in to place waste into appropriate compartments when residents hesitated or avoided handling bins. At the point of collection, workers described negotiating waste classification in real time, questioning households about contents and, in some cases, asking them to re-segregate to prevent mixing.

"We make them aware, and the siren in our vehicle is used to signal that wet and dry waste should be segregated. Cleanliness drives are conducted, and we also explain to households." (Datia: Service Provider (Frontline Sanitation Worker) IDI)

"Many people do this, they put it in themselves. The employee also contributes. Some people do not want to touch the dustbin, so they [staff] put it in themselves." (Deoria: Service Provider (Supervisor) IDI)

"If someone gives mixed waste, we sometimes stop them and ask them to segregate." (Datia: Service Provider (Frontline Sanitation Worker) IDI)

3.4.2 Organisational conditions shaping frontline mediation work

Service providers described mediation work as shaped by uneven training exposure, supervisory oversight, and competing duty allocations. Participants across cities reported variation in the duration, frequency, and formality of training, ranging from short, one-time sessions to periodic meetings, while some frontline workers reported receiving only basic or informal instruction. Supervisory monitoring was noted as an important mechanism for guiding behaviour in practice, often compensating for limited formal training.

"They were given a 5-day training, and they came from the department and gave the training. All the staff of a ward were called for training in one day." (Balasore: Service Provider (Local Authority) IDI)

"And we have only been provided with basic training, but if we talk about the training, it is needed in every field to improve our knowledge. (Datia: Service Provider (Frontline Sanitation Worker) FGD)

"No, there is no formal training." (Datia: Service Provider (Frontline Sanitation Worker) IDI)

At the same time, mediation work was described as occurring under tight temporal constraints, with workers reassigned across tasks or required to complete duties within compressed morning windows, limiting the time and attention available for household-level engagement.

"And an employee is working here, in the middle of that work, he is given the job of cleaning elsewhere. If his duty is in this area, he must see the management here and rather than send for work elsewhere." (Datia: Community Residents (Slum) Male FGD)

"We wake up at 3 am and go to work after doing all the housework. We have now been told to finish work between 6 and 9 in the morning due to the heat wave." (Balasore: Service Provider (Supervisor) FGD)

3.4.3 Stigma, recognition, and financial empowerment shaping sanitation workers' participation

Interactions at the point of waste handover were shaped by stigma and social hierarchies, which delineated clear boundaries between households and sanitation workers. Service providers described situations in which residents preferred to hand over waste themselves and explicitly resisted workers touching household dustbins or even approaching the home, indicating concerns about contamination and social distance. These interactional limits constrained the scope of worker mediation and shifted responsibility back to households, even when segregation practices were incomplete. Workers also noted that attempts to correct practices could be socially risky, as their lower perceived status reduced their ability to insist on compliance and exposed them to complaints or repercussions.

"Some people want [that] we don't even touch their dustbins, [they say] don't touch the bucket, don't even touch the gate". (Deoria: Service Provider (Supervisor) IDI).

"Now we people are little [low social standing], what can we say? If I tell, then they [Community Residents] will telephone so-and-so persons. And directly or indirectly, there is a problem. That will definitely come". (Deoria: Service Provider (Frontline Sanitation Worker) FGD)

While stigma constrained worker-household interactions, women sanitation workers in Balasore actively negotiated and, in some cases, reworked these boundaries through the visibility of

their public work. Participants noted that women who engaged in household waste segregation initially encountered shame, resistance, and moral scrutiny from community members and within their own families. Over time, however, sustained presence in ward-level cleaning and service delivery contributed to greater social acceptance, as residents began to associate women's sanitation work with tangible improvements in neighbourhood cleanliness. For the workers themselves, participation in sanitation work enabled new forms of mobility, income, and recognition beyond domestic roles, despite persistent hierarchies of respect.

"When women go to their house and segregate the garbage in front of them, they feel shame." (Balasore: Service Provider (Local Authority) IDI).

"I was not going out of the house, but I started this work after I went out, at first my mother-in-law opposed, but after my husband supported me, I joined this work." (Balasore: Service Provider (Supervisor) FGD)

"Yes, some people were protesting it. Some people were also commenting... how that woman is working in the garbage... Now they are not saying anything. Because they see that we are working in the Ward, that Ward is being cleansed." (Balasore: Service Provider (Supervisor) FGD)

"Yes, the drivers of BOV [Battery Operated Vehicle, meant for waste collection] are awarded in some functions" (Balasore: Service Provider (Frontline Sanitation Worker) FGD)

"Women don't get much respect, but they are very happy to get so much respect for the first time [as sanitation workers]." (Balasore: Service Provider (Supervisor) FGD)

3.4.4 Gendered structuring of frontline sanitation practices

Building on this gradual renegotiation of gendered stigma, women sanitation workers in Balasore consistently framed regular income as a central motivator sustaining their participation in sanitation work. Participants emphasised the importance of predictable, timely salaries, which contrasted sharply with earlier economic dependence within households. For many women, sanitation work enabled a reversal of financial dependency, allowing them to contribute to household expenses, support children's education, and exercise greater autonomy in everyday spending. This shift was especially salient during periods of economic disruption, such as the COVID-19 pandemic, when continued employment in sanitation work provided household stability. Service providers also described deliberately targeting economically vulnerable women for recruitment, positioning sanitation work as both livelihood support and service delivery.

"The biggest thing is that they get their salary by the 7th of every month regularly, for that they are interested in work." (Balasore: Service Provider (Local Authority) IDI)

"Earlier we had to ask others for our expenses, but now we are spending on ourselves and giving money to others to spend. We are

supporting our family for financial purposes... Spending on our own and saving some money for our future... If our baby needed anything, we could not give it; now we can give it all." (Balasore: Service Provider (Supervisor) FGD)

"During COVID, people were sitting at home, but we worked at that time, and I gave money to my husband, and with that money, my husband shopped. That's why we didn't have any problems during COVID." (Balasore: Service Provider (Supervisor) FGD)

"Rich women will not do this job, but we need to find needy women, they will work, those who need such work and money." (Balasore: Service Provider (Local Authority) IDI)

Both community residents and service providers described gender as actively shaping how sensitive sanitation interactions were mediated at the household level. Female sanitation workers were perceived as more appropriate interlocutors for waste practices involving bodily privacy, particularly sanitary waste, where embarrassment or social discomfort limited communication with male workers. Women participants emphasised mutual understanding and ease of interaction, noting that advice on disposal was more readily accepted when conveyed by women. Service providers similarly viewed women's involvement, especially through self-help groups, as enhancing trust and compliance during door-to-door engagement.

"Most probably are slum areas and we [males] don't say anything to them as the women give the garbage". (Deoria: Service Provider (Frontline Sanitation Worker) FGD)

"As the sweeper is a female, she can easily tell us how to dispose of the sanitary pad; it's different in case of a male to tell us regarding this... maybe the person felt embarrassed. It is useful for us". (Balasore: Community Residents (Non-slum) Female FGD)

Alongside this interactional advantage, sanitation work was also described as gender-differentiated in task allocation and skill acquisition. While frontline sanitation roles were often framed as compatible with women's domestic identities, participants also highlighted women's gradual acquisition of technical skills, including vehicle operation and campaigning, facilitated by on-the-job learning and institutional support.

"A man cannot do this work because he is doing service, but a woman can do this work and earn something because she is a housewife." (Balasore: Service Provider (Supervisor) FGD)

"We went to the municipality and learned to drive, and then we got the van. The male drivers in the municipality taught us how to drive". (Balasore: Service Provider (Frontline Sanitation Worker) FGD)

Both service providers and residents reported that male waste workers often face difficulties communicating with women about waste-related practices, highlighting the importance of engaging female waste management workers. Participants noted that because

women are primarily responsible for managing household waste, it is essential that they possess adequate waste segregation literacy, which female workers are perceived as communicating more effectively. The involvement of women as waste-related service providers was therefore viewed as beneficial for improving waste management practices, particularly in facilitating discussions around the disposal of sanitary and hazardous waste. Male service providers acknowledged discomfort in addressing these topics with women, expressing concern that such interactions could be misunderstood or perceived inappropriately.

"Most probably are slum areas and we [males] don't say anything to them as the women give the garbage". (FGD, Waste collecting vehicle driver – males, Deoria)

"As the sweeper is a female, she can easily tell us how to dispose of the sanitary pad; it's different in case of a male to tell us regarding this... maybe the person felt embarrassed. It is useful for us". (FGD, Non-slum female, Balasore)

3.5 Institutional embeddings and legitimacy flows

This theme explores how everyday segregation practices are shaped by how municipal systems, political influences, and communication processes generate, mediate, or undermine institutional legitimacy.

3.5.1 Post-collection opacity and the Erosion of institutional trust

Participants described limited visibility into what happened to waste after it was handed over, with post-collection processes largely opaque to households. While residents could sometimes identify nearby transfer points or trenching grounds, they reported little knowledge of subsequent handling, treatment, or recycling. Observations of waste dumped in the open, foul odours, and occasional burning near disposal sites contributed to perceptions that household-level segregation efforts may not translate into meaningful downstream processing. This lack of transparency weakened the perceived continuity of the waste management chain, positioning post-collection stages as distant, inaccessible, and beyond community influence. Respondents pointed to technological solutions, such as digital tracking systems, as a workaround.

"From here, the waste goes to the trenching ground. Is out of town. Which is on a XXXX road. We do not know what happens to the waste there. But when we pass by that road near the trenching ground, we see garbage left out in the open. Looks like a dumpyard. There is probably no recycling system there. Most of the garbage is set on fire there. When we cross that road, it stinks. This is the problem." (Datia: Community Residents (Slum) Male FGD)

"After giving this garbage, what is done with it next? Don't know about all this. Only the sweeper can tell where the garbage is taken from here, we do not know about this. (Deoria: Community Residents (Non-Slum) Female FGD)

3.5.2 Negotiating service legitimacy through user fees and sanctioning practices

User fees and challans were portrayed as instruments through which behavioural accountability around waste practices was expected to be reinforced. Penalties were framed not primarily as punitive, but as mechanisms to signal seriousness and induce responsibility after repeated non-compliance. Several providers argued that monetary payment, whether as a fine or a routine user charge, created a sense of obligation among residents to participate in segregation and proper handover.

"We tell them not to throw garbage outside. If he throws garbage outside without listening to us, we will let them go 2 times, and after 3rd time, we will give him notice to pay a penalty." (Balasore: Service Provider (Local Authority) IDI)

"If I am giving money monthly, so I will take care of the garbage that I have to give away. Only then will the public realise. We are failing because of the public not realising it." (Deoria: Service Provider (Supervisor) FGD)

At the same time, providers acknowledged tensions in applying penalties when supporting infrastructure or services were perceived as inadequate, suggesting that enforcement was often negotiated rather than absolute. In Balasore, disputes emerged where households engaging in alternative waste practices, such as diverting wet waste to livestock, questioned the basis for paying user fees at all. Community members also expressed scepticism about the authority and transparency of fee collection, particularly when receipts were perceived as informal or lacked government sanction. From the service-provider perspective, fee collection often became entangled with broader grievances about service quality, with inspectors describing how demands for payment were met with counter-claims about irregular cleaning or missed collection. In some cases, attempts to enforce fees were reported to have backfired, leading households to withdraw cooperation altogether and resume informal disposal.

"If you go to collect taxes, you will be told that cleaning is not being done and the garbage collector does not come every day. So, if we explain that the taxes we collect are not paid for cleanliness, then we have become their enemies." (Datia: Service Provider (Supervisor) IDI)

"At the back of this municipality, there is an old town where people keep cows, that's why they don't give wet garbage, and they also told us that we don't give garbage or user fees, but we told them if you throw out dry garbage, then you will be fined." (Balasore: Service Provider (Local Authority) IDI)

"No...no...They have asked me 3 times for the user fee. Also gave the receipt. I asked him if you are taking 30 rupees per month, what is the justification? He said we are taking the money and giving the receipt. I asked if the government issued the receipt or if you printed it [on your own]. What is the proof?" (Balasore: Community Resident (Non-Slum) Male FGD).

"We did that also (fee imposition) once, sir, but it was not successful. People stopped giving garbage and later started throwing it

here and there on the road". (Deoria: Service Provider (Frontline Sanitation Worker) IDI)

Alongside these frictions, service providers also described an incremental normalisation of user fees through their routinisation within everyday service delivery. In Balasore, user fees were framed less as a one-time compliance demand and more as a small, regular contribution integrated into door-to-door collection practices. Rather than universal acceptance, participants emphasised partial uptake over time, with repeated interaction, explanation, and visible service provision gradually increasing willingness to pay.

"People are paying us user fees of Rs 1 per day... They give monthly... Swachha supervisor collects user fee from house." (Balasore: Service Provider (Local Authority) IDI)

"People are saying that we are going to pay the holding tax, but why should we pay that tax? Still, some people have understood. It will take time to make all people understand... more than 50% are paying [now]." (Balasore: Service Provider (Supervisor) FGD)

3.5.3 Everyday uptake of segregation-related conventional communication

Service providers from Balasore and Deoria highlighted periodic, structured initiatives such as Swachhta Pakhwada campaigns and routine discussion forums (e.g., Swachha Charcha days) as formal channels through which guidelines, problems, and expectations were communicated. In contrast, some community participants from Datia reported little to no direct engagement focused on explaining segregation practices, and they experienced municipal presence primarily through cleaning activities rather than dialogue or instruction. Nevertheless, over time, participants also described a gradual accumulation of practical understanding of waste sorting, emerging from repeated exposure to messages, interactions, and routine practice rather than from single awareness events. Service providers noted that distinctions between wet, dry, and recyclable waste, previously unfamiliar to households, had become increasingly recognisable, alongside improved compliance with door-to-door collection.

"The situation was terrible earlier... Earlier, no one used to separate the wet and dry waste. Now, they separate the wet and dry waste. Now it is known where to put glass and where to put iron." (Datia: Service Provider (Supervisor) IDI, 230802_0219)

"We have been in this work for 3 years, and we have seen that people who did not give garbage before now have given garbage separately." (Balasore: Service Provider (Supervisor) FGD)

Both community members and service providers described a persistent gap between awareness and action, noting that knowledge of segregation rules did not consistently lead to behavioural change. Repeated campaigns, driver instructions, and supervisory messaging were often perceived as having diminishing effect, with segregation remaining sporadic and concentrated among a small subset of households. From the service provider perspective, this disconnect was sometimes framed as a need for compulsion or stronger enforcement

to convert awareness into routine practice, while community accounts suggested that being told or reminded alone was insufficient to disrupt established habits.

"More than 50% knows about it... but unless work is done forcefully, success is not achieved." (Deoria: Service Provider (Supervisor) FGD)

"Yes, sir, the driver tells us to put the garbage aside. But, no sir, no one does it with just that." (Deoria: Community Residents (Non-Slum) Male FGD)

While audio cues and slogans were often described as losing effectiveness over time, becoming background noise rather than prompts for action, awareness activities were characterised as shallow or infrequent. Some providers framed behavioural inertia as gradual and difficult to reverse, especially in the absence of tangible incentives, suggesting that communication alone had a limited impact on routine segregation practices.

"When the vehicle comes, the song also plays 'gaadi wala aaya ghar se kachra nikal' [The vehicle driver has come, handover the household waste]... But this song is not working." (Deoria: Community Residents (Non-Slum) Male FGD)

"Yes, but sometimes this kind of awareness is shallow. There is not enough awareness as much as it should be. They should come once or twice a month, but they come only when they need to." (Datia: Community Residents (Slum) Male FGD)

"We have already given it [leaflets], but whoever has a [negative] mentality cannot be changed. But we think that people will change slowly." (Balasore: Service Provider (Supervisor) FGD)

"The biggest problem is that now we cannot give them profits." (Deoria: Service Provider (Supervisor) FGD)

3.5.4 Adaptive reworking of communication practices to restore relevance

In response to perceived limits of routine messaging, service providers described ongoing experimentation with alternative, more immersive communication formats to re-engage communities. These efforts included rallies, door-to-door campaigns, street plays, wall paintings, LED displays, and ward-level initiatives conducted under programmes such as Swachhta Pakhwada and the Indian Swachhta League. Participants emphasised a shift toward visual and demonstrative approaches, particularly videos shown on mobile phones or televisions, or through ward-level screenings, to make waste flows and segregation processes more tangible to residents. Such strategies were framed as attempts to align communication with contemporary media consumption habits and to move beyond passive awareness toward more engaged participation by showing how household waste is handled after collection.

"Today's people do not listen to anything; they spend time only watching on TV and mobile. If we show documents and videos

through mobile and TV, then people will understand." (Balasore: Service provider (Frontline Sanitation Worker) FGD)

"[We are trying] ward-wise, to engage our community, sir, in this activity through a video. By showing the [waste management] video, when the garbage is collected and comes out here to be sorted, then what is the process here [municipal-level waste management], understand this process and be active and participate". (Deoria: Service Provider (Local Authority) IDI)

Participants also described other innovative means of engaging the community to achieve a sustainable solution. Service providers suggested efforts to make segregation salient by linking it to tangible local benefits, such as visible rewards for compliant households and clear explanations of health consequences, especially for children and older adults.

"We did ward campaigns, in which Swachha Sathi, supervisors, and sanitary inspectors went and explained to the people in schools and community centres. Those who are doing good work are being rewarded in ward campaigns." (Balasore: Service Provider (Local Authority) IDI)

"Like one thing is that you have ASHA workers, etc. Run one of your complex...campaign types. What diseases does it [waste] cause? [Tell them] children are affected, and old people are affected. When you know that this disease can be caused by this thing..." (Datia: Service Provider (Supervisor) IDI)

Participants emphasised schools as a long-term leverage point, arguing that sustained engagement with children could normalise segregation practices over time. Education was viewed as an investment whose effects would accumulate gradually, shaping future household routines through children's knowledge and influence within families.

"Education is our biggest long-term strategy to make people aware... if we explain the same thing to children in schools, then they can understand. And it would be very effective not today but after 10 years." (Deoria: Service Provider (Supervisor) FGD)

3.5.5 Hierarchical, socio-political, and administrative structuring of local waste governance

Participants described both close supervision of frontline workers through routine field checks, alongside limited engagement and recognition from higher administrative levels. This imbalance was perceived to create high expectations for performance without equivalent acknowledgement or support.

"The officers at the top do not look after the work of the lower level." (Deoria: Service Provider (Supervisor) FGD)

"There are 12 to 15 sweepers under the supervisor. We conduct field checks of every employee to see whether they are doing their job or not" (Deoria: Service Provider (Supervisor) IDI).

"It is unfortunate for this department that even if someone works, they are not given that much importance; there is a lot of expectation of work." (Deoria: Service Provider (Supervisor) FGD)

Service providers also noted that policy decisions regarding public-private partnership (PPP)-led models, outsourcing, and contract changes directly shaped everyday sanitation work. Such upstream administrative choices were experienced at the frontline as structural factors that influenced accountability, worker roles, and interactions with households.

"No, there is nothing like that [PPP model] right now. One year ago, the AVS company [a privately owned firm] came door-to-door for garbage collection and received a very good response, but the members and board decided to remove it at the time of the election. That company was doing good work, and the results were good." (Deoria: Service Provider (Supervisor) FGD)

Service providers described how everyday waste governance was shaped by informal political influence and social connections, which limited the ability of frontline workers to question mis-sorting or enforce norms. References to elected representatives, local power holders, or personal connections were perceived to discourage confrontation, particularly given workers' lower social standing and fear of repercussions. At the same time, participants also pointed to the constructive role of locally respected figures, such as councillors, social workers, teachers, and elders, whose involvement in ward committees or public events helped mobilise attention, legitimacy, and community cooperation around waste management practices.

"We can't argue [about waste mis-sorting]. What's the profit of arguing? Because they say that we know the Legislator and so and so person. All this is the problem here. Connections, too much of it [is used] here, For the maximum number of people here, I am telling the truth, people live upon such connections, even for small matters." (Deoria: Service Provider (Frontline Sanitation Worker) FGD)

"Even if we try to be a little strict, political manipulation happens. MLAs (Members of the Legislative Assembly) reach out for small matters" (Deoria: Service Provider (Supervisor) IDI)

"There are famous people here who are influencers people listen to. His role here is that whenever an event takes place, he is called, and people come together a little more, pay attention, get motivated, and listen to him. They are a good social worker, teacher, and elderly people." (Datia: Service Provider (Frontline Sanitation Worker) FGD)

3.5.6 Felt need and efforts to modify material and manpower resource infrastructure

Service providers described segregation and processing as being closely tied to existing material capacity, including the availability of centralised processing facilities, MRFs, composting facilities, machinery, and adequate space. Shortfalls in equipment and specialised facilities, particularly for sanitary waste, limited the extent to which waste

could be processed as intended. Although incremental investments and planned centralised plants were noted, participants emphasised that current infrastructure largely set the operational ceiling for segregation outcomes.

"We don't have any machines for burning the sanitary waste". (Balasore: Service Provider (Frontline Sanitation Worker) FGD)

"We built an MRF... prepared a wet compost plant... we do the work by hand, but a lot of garbage is what we cannot do by hand... To increase the capacity, more new projects and machines have also been installed." (Datia: Service Provider (Supervisor) IDI)

"Look at the MRF Centre, the more the door-to-door garbage increases, the more space we [would] need." (Deoria: Service Provider (Frontline Sanitation Worker) FGD)

Linked to these infrastructure constraints, respondents noted that limited vehicle fleets and uneven ward coverage shaped how collection was organised. Insufficient numbers of vehicles required mixed modalities, combining door-to-door collection with handcarts or main-road pickup, which reduced service delivery consistency and affected households' ability to participate reliably in segregation.

"But the problem is that we don't have that many resources. There are 25 vehicles in 25 wards, but we need at least four vehicles in a ward." (Deoria: Service Provider (Supervisor) FGD)

Together, constrained infrastructure and limited vehicle capacity resulted in large volumes of mixed waste reaching processing sites, where manual labour became the primary means of secondary sorting. Participants described increased staffing requirements, longer work hours, and incomplete recovery of materials, indicating that labour-intensive practices functioned as a compensatory mechanism within an otherwise capacity-limited system.

"We need free helping hands so that everything can be done on time." (Balasore: Service Provider (Supervisor) FGD)

"...and now we need to engage 30 people instead of 10 people in the plant. If we have more manpower, they will work in shifts, so that people do not have to work hard and work will [still] continue throughout the day." (Balasore: Service Provider (Local Authority) IDI)

"As long as the garbage is not coming separated and people are not giving it separately, there is a need for more employees here [MRF Centre]". (Deoria: Service Provider (Frontline Sanitation Worker) FGD)

3.5.7 Sustained effort, local adaptation, and interactional adjustments in community engagement

This sub-theme compiles participants' perspectives on frontline waste governance and segregation, which is described as an accumulative, practice-based process shaped by sustained

engagement, everyday negotiation, and adaptive problem-solving. Repeated house-to-house interactions were perceived to gradually stabilise segregation practices, with workers spending extended time in wards to explain, remind, and normalise expectations. Alongside this, service providers reported developing situated workarounds to manage spatial, temporal, and household constraints, such as collecting waste from doorsteps, upper floors, or narrow lanes, and flexibly adjusting to mismatches between household disposal rhythms and collection schedules. These practices were underpinned by interactional norms that emphasised respectful conduct by workers, recognised gendered household contexts, and emphasised polite, relational engagement. Participants also described symbolic reframing of waste spaces, such as renaming or beautifying facilities, as a strategy to reduce stigma and reshape public perception of waste-related sites. Finally, existing complaint and feedback mechanisms, including helplines and informal reporting pathways, were viewed as routinised channels through which communities could signal problems and trigger corrective action.

In addition to the thematic patterns described above, city-level comparisons revealed important differences in how procedural clarity and service organisation shaped how waste segregation was understood and enacted. Balasore stood out for its relatively more differentiated service design, including clearer compartmentalisation of vehicles and reports of designated collection arrangements for specific waste streams, particularly hazardous waste. This was accompanied by a deliberate institutional choice to rely on women's self-help groups rather than private contractors, with service providers emphasising the motivational and communicative advantages of peer-to-peer engagement by women sanitation workers. In contrast, Datia and Deoria exhibited greater procedural ambiguity at the household-collection interface. Participants in both cities described inconsistent vehicle designs, limited explanation of waste categories, and uneven enforcement of collection routines, which collectively contributed to momentary, on-the-spot sorting decisions during handover. While both cities had experimented with or supported private collection models, these arrangements were described as politically contingent and unevenly sustained, leading to variability in service continuity and public expectations. Importantly, differences were not only inter-city but also intra-city, with contrasting views among service providers and residents regarding privatisation, enforcement, and responsibility.

4 Discussion

This study provides a practice-based explanation for why household waste segregation remains fragile in small Indian cities despite clear policy intent and widespread awareness. Rather than locating non-segregation primarily in individual deficits of knowledge or motivation, our findings indicate that segregation outcomes are produced through the interaction of temporal synchronisation constraints, lived material infrastructures, and legitimacy and mediation dynamics embedded in everyday service delivery. Across the three study sites, segregation did not operate as a stable household behaviour but rather as a situational accomplishment continuously negotiated at the interface between household routines, frontline collection practices, and local institutional arrangements.

This practice-oriented framing complements broader accounts of municipal solid waste management challenges in low- and middle-income settings by clarifying how routine implementation frictions accumulate at the doorstep and along the chain, particularly outside large metropolitan contexts (Zhang et al., 2024; Kosoe, 2024) (see Figure 2).

A central finding of this study is that waste segregation is highly sensitive to temporal alignment between household routines and collection practices. Morning collection schedules, while administratively fixed, intersected unpredictably with household rhythms shaped by bathing, eating, work departures, and school timings. As documented in the results, missed handovers were frequently not due to refusal or disinterest, but to tight time budgets, embodied norms around cleanliness after bathing, informal morning attire, and gendered domestic labour that concentrates early-morning coordination work among women. These findings align with evidence from small-town and developing-country settings showing that collection performance and participation are shaped by operational design and route-level constraints, rather than by household-level awareness alone (Kumar K. et al., 2024; Del Carmen-Niño et al., 2023; Yukalang et al., 2017).

These observations complicate interpretations, based on the existing literature, that frame non-segregation primarily as wilful non-compliance or low awareness (Sahoo et al., 2024). Instead, they show how segregation becomes fragile when timing uncertainty and fast-paced interactions reduce households' capacity to enact prescribed practices. Even among residents who articulated segregation norms, the inability to synchronise with collection rhythms led to practical compromises, particularly during brief vehicle halts and compressed handover windows. The implication is not that households are unmotivated, but that service designs often externalise coordination costs onto residents, making consistent enactment difficult in everyday life. This is consistent with broader findings that household separation intentions and reported practices are contingent on feasibility and perceived system functionality, not solely on individual attitudes (Nguyen et al., 2015; Yukalang et al., 2017; Sahoo et al., 2022).

Beyond time, segregation practices were shaped by how material infrastructures signalled, enabled, or undermined sorting expectations. Households commonly organised waste using single or limited bin arrangements, not necessarily due to ignorance of guidelines, but because storage practices were calibrated to what collection systems appeared to support. Where vehicles arrived with a single compartment, or where mixed and segregated vehicles coexisted within the same city, households questioned the need to maintain separation at home. This extends prior work on waste separation behaviour by specifying a practical mechanism, namely how system cues and perceived downstream handling influence whether households see segregation as meaningful or performative (Yukalang et al., 2017; Sahoo et al., 2022).

Procedural ambiguities further compounded this problem. Participants repeatedly described uncertainty about how specific materials, particularly plastic bags containing wet waste and certain sanitary items, should be classified. This finding aligns with our previous research, which has highlighted the need to standardise waste categorisation and associated terminology (Krishnan et al., 2026). Respondents pointed out that vehicle labels were rarely accompanied by explanations, leaving them to rely on inference or improvisation during source segregation. The downstream consequences described by service providers in our study are consistent with the wider risk

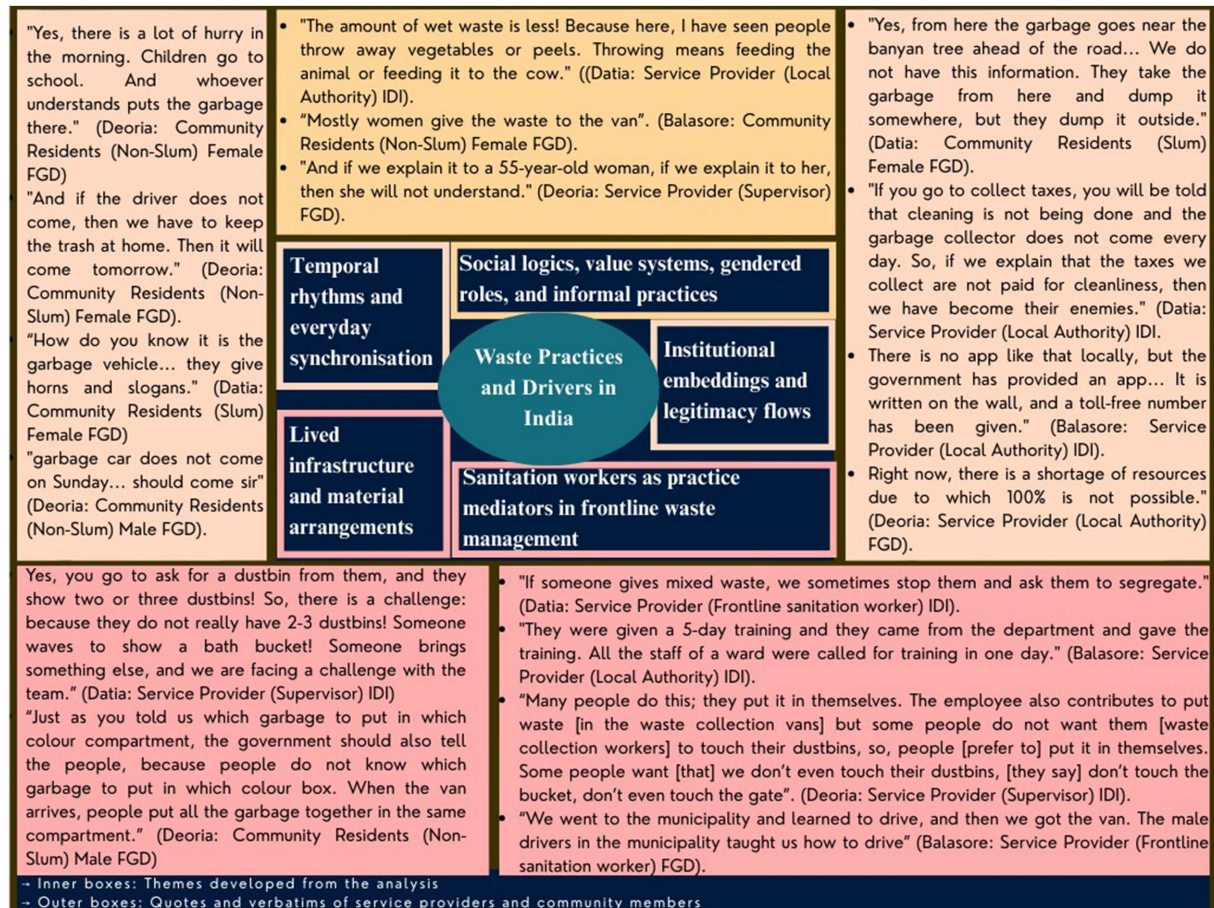


FIGURE 2

Important quotes highlighting the interaction of temporal synchronisation constraints, lived material infrastructures, and legitimacy and mediation dynamics in waste management in small Indian cities.

literature, which emphasises that mismanagement and mixing can increase health and environmental risks across disposal sites, processing points, and the broader environment (Yuan and Zoungana, 2025; De Titto and Savino, 2024; Njewa et al., 2025).

Household waste practices were also shaped by social meanings and value systems that do not map neatly onto formal policy categories. Participants consistently framed segregation in terms of everyday effects, such as reducing odour, preventing disease, protecting animals, and maintaining social respectability. These care-based motivations often supported segregation but also led to selective adaptations. For instance, the practice of placing wet waste in plastic bags reflected an effort to keep household bins clean and odour-free, even though it conflicted with official segregation rules. Such accounts resonate with people-planet-profit framings in food and organic waste management, where household practices around leftover food reuse and organic waste diversion are shaped by perceived utility, cost savings, and reduced waste volume, even when these practices operate outside formal municipal systems (Parsa et al., 2024).

In similar lines, respondents' descriptions underscore how informal diversion of waste, including feeding wet waste to livestock, household composting, and selling recyclables to scrap dealers, structured what entered the municipal waste stream in the first place. Their accounts show that these practices were not residual or marginal but integral to how households reasoned about waste value and responsibility, with the practices themselves shaped by perceived feasibility and local context. This aligns with comparable practices observed

elsewhere in India, where cattle feeding on vegetable and organic waste reduced waste (Alam et al., 2023). At the same time, diversion to live-stock and associated animal-health concerns highlight why informal organic flows can create both opportunities and risks. Where plastic ingestion by cattle has been documented, the issue is not merely "litter," but a welfare and public-health externality that small-city systems may face more visibly due to close human-animal co-presence and open-access waste environments (Ashok et al., 2021; Sahu et al., 2022).

Given that organic waste constitutes a substantial proportion of municipal waste in developing countries (Njewa et al., 2025; Jalalipour et al., 2024), these findings suggest, from a policy perspective, that informal organic diversion should not be treated as non-compliance, but as an existing waste ecology that could be incorporated more deliberately, with safeguards. Additionally, such practices are expected to be especially prominent in smaller cities compared to larger metropolitan areas, owing to lower urban live-stock presence in the latter (Lindahl et al., 2018). Thus, at a policy scale, it would be prudent to explore safe means to strategically leverage such existing avenues to reduce landfill pressure without relying solely on cost-intensive composting infrastructure (Mulyana et al., 2024). However, incorporation would require clear guidance on what is safe to divert, how to prevent plastic contamination of feed/manure streams, and how municipal systems can support or certify these practices without shifting risk onto households. Therefore, further research is needed to expand on the ongoing work exploring ways to mitigate the hazards linked to close human-animal

co-presence (Ankur, 2023). In this sense, our findings align with the broader literature that frames peri-urban and small-city waste management as an environmental health challenge, where solutions must explicitly manage risk pathways rather than simply increase collection volumes (Kumar G. et al., 2024; Yuan and Zoungana, 2025; De Titto and Savino, 2024; Njewa et al., 2025).

A key finding of this study is the documentation of evidence foregrounding sanitation workers as active mediators who co-produce segregation practices at the doorstep. Workers routinely engaged in real-time classification, explanation, and corrective action, often compensating for household uncertainty or reluctance. This extends our prior qualitative work showing that segregation behaviour is dynamic and relational, shaped by ongoing interaction between households and frontline systems rather than by one-time behaviour change (Sahoo et al., 2024; Sahoo et al., 2022). However, mediation capacity was shaped by organisational conditions, including uneven training, time pressures, and reassignment across duties. Social hierarchies and stigma further constrained mediation work, as workers described limits on their ability to question households, particularly where residents invoked social or political connections. These constraints are consistent with broader work highlighting that trust, perceived fairness, and social justice dynamics influence household participation, and that system legitimacy can be undermined when interactions are experienced as coercive, arbitrary, or socially unequal (Veliverronena and Davidstone, 2025).

At the same time, our findings from Balasore indicate that the gendered structuring of frontline sanitation work can operate as an implementation asset rather than merely a social descriptor. Women sanitation workers and self-help groups appeared to enhance communication around sensitive waste streams, reduce embarrassment, and foster trust, while also providing pathways for recognition and financial empowerment. This is consistent with policy and programme discussions that emphasise gender inclusion and women-led sanitation initiatives, and with evidence that gender-inclusive approaches can strengthen waste segregation outcomes when designed as an operational strategy rather than a symbolic add-on (United Nations Environment Programme International Environment Technology Centre, 2022; Ministry of Housing and Urban Affairs, 2023).

Segregation practices were deeply shaped by perceptions of institutional legitimacy. A recurrent theme across sites was post-collection opacity; households had little visibility into what happened to waste after handover, and observations of open dumping or burning undermined confidence that segregation efforts mattered. This legitimacy problem directly connects to prior evidence that dumping sites and mismanagement contribute to environmental contamination and risk, which, in turn, can erode community trust in the value of household-level effort (Yuan and Zoungana, 2025; De Titto and Savino, 2024; Njewa et al., 2025). Rebuilding legitimacy would depend on making waste flows visible and auditable, and on closing feedback loops between effort and outcome. Previous literature has shown that digital and sensor-enabled systems, including RFID-enabled tracking and related smart-waste approaches, may help address visibility and accountability gaps, particularly around route adherence and collection verification (Kumar G. et al., 2024; Ranjana and Eliyas, 2021; Gautam and Singh, 2022). However, the implementation literature also cautions that digital solutions can be difficult to sustain without parallel investment in maintenance capacity, worker training, and institutional accountability, particularly in smaller municipalities

with limited technical staffing and budgets (Atofarati et al., 2025; Das, 2025; Hassan, 2025).

User fees and penalties were similarly ambivalent instruments. While some service providers viewed fees as necessary signals of seriousness, their implementation was frequently contested when service quality was perceived as uneven or when households engaged in alternative disposal practices. This mirrors broader debates on pay-as-you-throw and willingness-to-pay, where acceptance depends on perceived service reliability, fairness, and transparency, and where poorly aligned enforcement can trigger resistance rather than compliance (Romano and Masserini, 2023; Nguyen et al., 2015; Ali et al., 2025; Rai et al., 2019). Our findings suggest that legitimacy is not secured through policy instruments alone, but through credible feedback loops that connect household effort to observable outcomes, and that align charging and enforcement with demonstrably reliable service delivery.

Communication campaigns showed diminishing returns when detached from such feedback. Repeated slogans and audio cues were often described as ineffective, background noise, while more immersive and demonstrative strategies were perceived as more promising. This aligns with evidence supporting intergenerational and education-based approaches, in which children and schools can shape longer-run household practices, and with evidence reviews showing that education and structured engagement can improve awareness and behaviour beyond one-off messaging (Deng et al., 2022; Maddox et al., 2011; Awino and Apitz, 2024; Debrah and Dinis, 2021). Media and communication approaches that are demonstrative rather than purely exhortative may also be relevant, particularly where they make downstream processes visible and credible to communities (Herdiansyah et al., 2021).

This study has several limitations. As a qualitative investigation, the findings reflect issues that emerged as salient to participants during interviews and FGDs. The absence of a particular theme in discussions within a given city should not be interpreted as the absence or irrelevance of that issue in that context, but rather as an indication that it did not emerge as a priority relative to other concerns. City-wise differences should therefore be understood as reflecting differential stakeholder prioritisation shaped by local context and lived experience, rather than definitive presence or absence of structural conditions. In addition, the findings may not be generalisable to all small cities, as cultural norms, governance structures, and resource availability vary considerably. Participant responses may also have been influenced by social desirability bias, particularly in group discussions. The cross-sectional design limits insights into how perceptions and practices evolve. Finally, the exclusion of private-sector actors and senior municipal decision-makers may have constrained analysis of higher-level planning, financing, and policy enforcement dynamics.

5 Conclusion

Overall, the findings indicate that improving household waste segregation in small cities requires moving beyond awareness-centric or demographic targeting approaches to address how segregation is practically organised, signalled, and sustained in everyday life. Effective interventions must align collection schedules and routing with household rhythms,

strengthen reliability at the point of handover, and ensure clarity and consistency in vehicle design and waste category communication to reduce conditional or symbolic compliance. Rather than treating informal diversion practices as failures of the system, recognising them as part of the existing waste ecology may offer opportunities to build policy around locally embedded logics, particularly for organic waste, while managing associated environmental and public-health risks. Similarly, stabilising segregation practices depends on strengthening frontline mediation through adequate staffing, training, and workforce design that leverages gendered interactional advantages and provides recognition and security. Finally, institutional legitimacy emerges as central to sustained participation. Making waste flows more visible and closing feedback loops between household effort and downstream outcomes are critical for maintaining trust and compliance. While digital and smart-waste solutions may help address opacity and accountability gaps, their effectiveness in small cities ultimately depends on parallel investments in maintenance capacity, worker support, and institutional oversight. Together, these insights underscore that persistent gaps between policy intent and everyday practice arise not only from a lack of awareness but also from misalignments between lived routines, material systems, and governance arrangements that shape waste management in small-city contexts.

Data availability statement

The original contributions presented in the study are included in the article/supplementary material, further inquiries can be directed to the corresponding author.

Ethics statement

The studies involving humans were approved by Institutional Ethics Committee of ICMR-National Institute for Research in Environmental Health (NIREH), Bhopal. The studies were conducted in accordance with the local legislation and institutional requirements. Written informed consent for participation in this study was provided by the participants' legal guardians/next of kin.

Author contributions

TT: Conceptualization, Methodology, Project administration, Visualization, Writing – original draft, Formal analysis, Validation. MY: Formal analysis, Visualization, Writing – original draft, Methodology. KS: Conceptualization, Methodology, Data curation, Formal analysis, Writing – review & editing. SS: Conceptualization, Methodology, Writing – review & editing, Project administration, Visualization. VP: Data curation, Methodology, Visualization, Writing – review & editing, Project administration. AP: Data curation, Methodology, Resources, Writing – review & editing, Investigation, Project administration. YS: Conceptualization, Methodology, Resources, Supervision, Writing – review & editing, Investigation. SP: Resources, Supervision, Validation, Writing – review & editing. RT: Methodology, Resources, Writing – review & editing, Validation. VD: Conceptualization, Data curation, Formal analysis, Funding

acquisition, Methodology, Project administration, Resources, Supervision, Writing – review & editing, Investigation, Visualization.

Funding

The author(s) declared that financial support was received for this work and/or its publication. The work was supported by the Indian Council of Medical Research vide Sanction Letter No ICMR-NIREH/BPL/IMP-PH49/2022-23/96 dated 22/03/2023 issued to PI (Dr. VD) at ICMR-NIREH, Bhopal. The funders had no role in study design, data collection and analysis, decision to publish, or preparation of the manuscript.

Acknowledgments

The authors are grateful to all study participants for their time and valuable insights. We thank the field staff for their support in data collection, translation, and transcription. We also acknowledge the Municipalities of Balasore, Datia, and Deoria for granting permission to conduct the study. We sincerely thank Dr. Madhanraj K. and Ms. Kavya Krishnan for their initial inputs on the study objectives; Mr. Rajesh Rathore for supporting FGD/IDI translations; and Mr. Atul Trivedi, Divisional WASH Consultant, SBM, Gwalior Division, MP, for his valuable support in facilitating local coordination and community engagement in Datia.

Conflict of interest

The author(s) declared that this work was conducted in the absence of any commercial or financial relationships that could be construed as a potential conflict of interest.

Generative AI statement

The author(s) declared that Generative AI was not used in the creation of this manuscript.

Any alternative text (alt text) provided alongside figures in this article has been generated by Frontiers with the support of artificial intelligence and reasonable efforts have been made to ensure accuracy, including review by the authors wherever possible. If you identify any issues, please contact us.

Publisher's note

All claims expressed in this article are solely those of the authors and do not necessarily represent those of their affiliated organizations, or those of the publisher, the editors and the reviewers. Any product that may be evaluated in this article, or claim that may be made by its manufacturer, is not guaranteed or endorsed by the publisher.

References

- Ahluwalia, I. J., and Patel, U. (2018). Solid waste management in India: an assessment of resource recovery and environmental impact. Working Paper No. 356. New Delhi: Indian Council for Research on International Economic Relations (ICRIER). Available online at: <https://icrier.org/publications/solid-waste-management-in-india-an-assessment-of-resource-recovery-and-environmental-impact/> (Accessed November 24, 2025).
- Alam, M. S. V. S., Dey, D. K., Adilieme, C., Malik, P. K., Bhatta, R., König, S., et al. (2023). Urbanisation threats to dairy cattle health: insights from greater Bengaluru, India. *Trop. Anim. Health Prod.* 55:350. doi: 10.1007/s11250-023-03737-7
- Ali, S., Akter, S., Boros, A., and Temesi, I. (2025). Factors influencing households' willingness to pay for advanced waste management services in an emerging nation. *Urban Sci.* 9:270. doi: 10.3390/urbansci9070270
- Ankur, K. R. a. (2023). Quality circle approach equipped with AI to solve the issue of stray cattle detection and identification in small cities to avoid accidents. *Int. J. Multidiscip. Res.* 5. doi: 10.36948/ijfmr.2023.v05i04.5112
- Ashok, K., Babu, M., Jula, V., and Mullai, N. K. (2021). Digestive impact of cow when ingest plastic waste. *Linguist. Cult. Rev.* 5, 1257–1264. doi: 10.21744/lingure.v5ns1.1597
- Atofarati, E. O., Adogbeji, V. O., and Enweremadu, C. C. (2025). Sustainable smart waste management solutions for rapidly urbanizing African cities. *Util. Policy* 95:101961. doi: 10.1016/j.jup.2025.101961
- Awino, F. B., and Apitz, S. E. (2024). Solid waste management in the context of the waste hierarchy and circular economy frameworks: an international critical review. *Integr. Environ. Assess. Manag.* 20, 9–35.
- Baleswar Action Plan. District environment plan, Baleswar, Odisha. (2023). Waste-Wise Cities: Best practices in municipal solid waste management. NITI Aayog, Government of India. Available online at: <https://balasore.odisha.gov.in/sites/default/files/2023-08/2023012112.pdf> (Accessed 5 December 2025).
- Biswas, A., Chaudhary, K., Singh, R., Tewari, S., Singh, S., and Parida, S. (2021). Available online at: <https://www.niti.gov.in/sites/default/files/2021-12/Waste-Wise-Cities.pdf>. [Accessed on: 24 November 2025].
- Census of India-2011. District Census handbook-Datia. (2015). Available online at: https://censusindia.gov.in/nada/index.php/catalog/697/download/2436/DH_2011_2305_PART_B_DCHB_DATIA.pdf (Accessed 5 December 2025).
- Census-of-India-2001. District Census Handbook Part—A & B. (2014). Available online at: https://censusindia.gov.in/nada/index.php/catalog/27749/download/30918/DH_23_2001_DAT.pdf (Accessed 5 December 2025).
- Das, D. K. (2025). Digital technology and AI for smart sustainable cities in the global south: a critical review of literature and case studies. *Urban Sci.* 9:72. doi: 10.3390/urbansci9030072
- Datia Environment Plan. Datia district action plan. (2021). Available online at: <https://www.mppcb.mp.gov.in/proc/Tech/DEP%20Datia.pdf> (Accessed 5 December 2025).
- De Titto, E., and Savino, A. (2024). Human health impact of municipal solid waste mismanagement: a review. *Adv. Environ. Eng. Res.* 5, 1–37. doi: 10.21926/aer.2402014
- Debrah, J. K. V. D., and Dinis, M. A. (2021). Raising awareness on solid waste management through formal education for sustainability: a developing countries evidence review. *Recycling* 6:6. doi: 10.3390/recycling6010006
- Del Carmen-Niño, V., Herrera-Navarrete, R., Juárez-López, A. L., Sampedro-Rosas, M. L., and Reyes-Umaña, M. (2023). Municipal solid waste collection: challenges, strategies and perspectives in the optimization of a municipal route in a southern Mexican town. *Sustainability* 15:1083. doi: 10.3390/su15021083
- Deng, J., Tang, J., Lu, C., Han, B., and Liu, P. (2022). Commitment and intergenerational influence: a field study on the role of children in promoting recycling in the family. *Res. Conserv. Recycl.* 185:106403. doi: 10.1016/j.resconrec.2022.106403
- Dhanuskar, S., Naik, S., and Pant, K. (2024). "Environmental impact of waste to energy" in Biodegradable waste processing for sustainable developments, 51. Prasad A., and Paul AK, eds. CRC Press (Taylor & Francis Group), 51–86. doi: 10.1201/9781003502012-3
- Frazer-Williams, R., and Sankoh, A. (2024). "Soil contamination resulting from inefficient solid waste management" in *Environmental pollution and public health* Case studies on air, water and soil from an interdisciplinary perspective R. Frazer-Williams, M. B. Ogundiran, and E. I. Unuabonah eds. (Amsterdam, The Netherlands: Elsevier), 251–264. doi: 10.1016/C2021-0-02196-9
- Gautam, H. C. Y. V., and Singh, V. (2022). *IoT-enabled Services for Sustainable Municipal Solid Waste Management in India*. 1st Edn: Boca Raton 2022.
- Government of Odisha. Balasore: about us 2025. Available online at: <https://balasore.odisha.gov.in/about-district/about-us#:~:text=Balasore%20District%20covers%20an%20area,lies%20on%20its%20western%20side> (Accessed November 24, 2025).
- Hamdan, A., Panda, S., Jain, M. S., Raj, V., and Mathew, S. (2025). Assessing municipal solid waste in Indian smart cities: a path towards waste-to-energy. *Heliyon* 11. doi: 10.1016/j.heliyon.2025.e42770
- Hassan, O. M. (2025). Integrated digital, biological, and human capital innovations for circular and sustainable waste management: a critical review. *Discover Applied Sciences*. 7, 1–21. doi: 10.1007/s42452-025-07904-3
- Herdiansyah, H. B. A., Negoro, H. A., Sari, R., and Zakianis, Z. (2021). Parental education and good child habits to encourage sustainable littering behavior. *Sustainability* 13:8645. doi: 10.3390/su13158645
- International Trade Administration. India Solid Waste Management. Washington (DC): U.S. Department of Commerce; (2023). Available online at: <https://www.trade.gov/market-intelligence/india-solid-waste-management> (Accessed 6 June).
- Jalalipour, H., Binaee Haghighi, A., Ferronato, N., Bottausci, S., Bonoli, A., and Nelles, M. (2024). Social, economic and environmental benefits of organic waste home composting in Iran. *Waste Manag. Res.* 43:0734242X241227377. doi: 10.1177/0734242X241227377
- Kalyanasundaram, M., Krishnan, K., Singh, S., Sahoo, K. C., Soni, R., Parashar, V., et al. (2023). Composition analysis (pick analysis) of waste generated from household: a pilot study in Ujjain city, India. *Heliyon* 9:e19902. doi: 10.1016/j.heliyon.2023.e19902
- Kosoe, E. A. (2024). Solid waste management practices in small towns: implications for urban sustainability in Ghana. *Sci. Dev. J.* 8:29. Available online at: https://www.researchgate.net/publication/382428585_Solid_Waste_Management_practices_in_Small_towns_Implications_for_Urban_sustainability_in_Ghana
- Krishnan, K., Singh, S., Trushna, T., et al. (2026). Standardizing waste classification in low and middle income countries: addressing terminology gaps to improve urban segregation behavior. *Int. J. Environ. Sci. Technol.* 23:185. doi: 10.1007/s13762-025-06994-1
- Kumar, K., Kumar, R., and Kumar, A. (2024). Assessment of door-to-door waste collection in Solan town of Himachal Pradesh. *Soc. Work* 13, 118–141. Available online at: https://www.researchgate.net/profile/Rohit-Kumar-557/publication/383669511_Assessment_of_Door-toDoorWasteCollectioninSolanTownofHimachalPradesh/links/66d6a9b664f7bf7b194f84f7/Assessment-of-Door-to-Door-Waste-Collection-in-Solan-Town-of-Himachal-Prp.pdf, accessed on 06-02-2026
- Kumar, G., Vyas, S., Sharma, S. N., and Dehalwar, K. (2024). "Challenges of environmental health in waste Management for Peri-urban Areas" in *Solid waste management: Advances and trends to tackle the SDGs* (Springer), 149–168.
- Leknoi, U., Yienthaisong, A., and Likitlersuang, S. (2024). Social factors influencing waste separation behaviour among the multi-class residents in a megacity: a survey analysis from a community in Bangkok, Thailand. *Sustain. Fut.* 7:100202. doi: 10.1016/j.sfr.2024.100202
- Lenkiewicz, Z. UNEP global waste management outlook 2024: beyond an age of waste-turning rubbish into a resource. Nairobi; (2024). Available online at: https://wedocs.unep.org/bitstream/handle/20.500.11822/44939/global_waste_management_outlook_2024.pdf?sequence=3 (Accessed 12 December 2025).
- Lindahl, J. F., Magnusson, U., and Grace, D. (2018). "Urban livestock-keeping: contributions to food and nutrition security" in Encyclopedia of food security and sustainability. P. Ferranti, E. M. Berry, & J. R. Anderson (eds.), 317–321. Elsevier. Available online at: <https://doi.org/10.1016/B978-0-08-100596-5.21531-X>
- Maddox, P., Doran, C., Williams, I. D., and Kus, M. (2011). The role of intergenerational influence in waste education programmes: the THAW project. *Waste Manag.* 31, 2590–2600. doi: 10.1016/j.wasman.2011.07.023
- Ministry of Housing and Urban Affairs. (2023). A Swachh Bharat Mission—urban 2.0 initiative, government of India. Women-Led Sanitation: Stories of change 2023. India. Available online at: <https://sbmurban.org/storage/app/media/stories-of-change.pdf> (Accessed 13 July 2025).
- Mondal, S., and Mandal, B. (2024). Assessment of urban solid waste management in a class II Indian city using geospatial and statistical approaches: a case study of Rampurhat municipality. *Waste Manag. Bull.* 1, 74–92. doi: 10.1016/j.wmb.2023.09.003
- Mulyana, R., Suryawan, I. W. K., Ulhasanah, N., Septiariva, I. Y., Prayogo, W., Suhardono, S., et al. (2024). Nature-based solutions in urban landscapes: determinants influencing willingness to participate in composting in metropolitan Jakarta, Indonesia. *J. Nat. Res. Environ. Manag.* 14:451. doi: 10.29244/jpsl.14.3.451
- Municipal Solid Wastes. (2023). Available online at: [https://mohua.gov.in/upload/upload-files/files/Part1\(1\).pdf](https://mohua.gov.in/upload/upload-files/files/Part1(1).pdf) (Accessed on: 12 August 2025)
- Nguyen, T. T. P., Zhu, D., and Le, N. P. (2015). Factors influencing waste separation intention of residential households in a developing country: evidence from Hanoi, Vietnam. *Habitat Int.* 48, 169–176. doi: 10.1016/j.habitatint.2015.03.013
- Njewa, J. B., Mweta, G., Sumani, J., and Biswick, T. T. (2025). The impact of dumping sites on air, soil and water pollution in selected southern African countries: challenges and recommendations. *Water Emerg. Contamin. Nanopl.* 4:3. doi: 10.20517/wecn.2024.71
- Parsa, A., Van De Wiel, M., Schmutz, U., Taylor, I., and Fried, J. (2024). Balancing people, planet, and profit in urban food waste management. *Sustain. Prod. Consum.* 45, 203–215. doi: 10.1016/j.spc.2024.01.003
- Rai, R. K., Bhattarai, D., and Neupane, S. (2019). Designing solid waste collection strategy in small municipalities of developing countries using choice experiment. *J. Urban Manag.* 8, 386–395. Elsevier. doi: 10.1016/j.jum.2019.07.001. (Accessed November 24, 2025).
- Rai, S., Gurung, A., Sharma, H. B., Ranjan, V. P., and Cheela, V. R. S. (2024). Sustainable solid waste management challenges in hill cities of developing countries: insights from eastern Himalayan smart cities of Sikkim, India. *Waste Manag. Bull.* 2, 1–18. doi: 10.1016/j.wmb.2024.02.009
- Ranjana, P. V. S., and Eliyas, S. (2021). IoT based smart garbage collection using RFID and sensors. *J. Phys. Conf. Ser.* 1818:012225. doi: 10.1088/1742-6596/1818/1/012225

- Romano, G., and Masserini, L. (2023). Pay-as-you-throw tariff and sustainable urban waste management: an empirical analysis of relevant effects. *J. Environ. Manag.* 347:119211. doi: 10.1016/j.jenvman.2023.119211
- Sahoo, K. C., Kalyanasundaram, M., Singh, S., Soni, R., Parashar, V., Pathak, A., et al. (2024). Service providers and program managers' perspectives on household waste segregation in Ujjain, India. *Sustain. Environ.* 10:2383472. doi: 10.1080/27658511.2024.2383472
- Sahoo, K. C., Soni, R., Kalyanasundaram, M., Singh, S., Parashar, V., Pathak, A., et al. (2022). Dynamics of household waste segregation behaviour in urban community in Ujjain, India: a framework analysis. *Int. J. Environ. Res. Public Health* 19:7321. doi: 10.3390/ijerph19127321
- Sahu, B. K., Parganiha, A., and Pati, A. K. (2022). Time-of-day and seasonal variations in foraging behavior of street cattle of urban Raipur, India. *Biol. Rhythm. Res.* 53, 786–800. doi: 10.1080/09291016.2020.1794646
- Sarker, A., Baul, T. K., Nath, T. K., Karmakar, S., and Paul, A. (2024). Household solid waste management in a recently established municipality of Bangladesh: prevailing practices, residents' perceptions, attitude and awareness. *World Dev. Sustain.* 4:100120. doi: 10.1016/j.wds.2023.100120
- Shukla, N., Mallick, R., Chakraborty, D., and Gupta, S. (2025). *Public health implication of toxic waste generated by households: The environmental hazards of improper waste disposal. producing healthy food with healthy soils: Hazardous waste management.* Cham, Switzerland: Springer, 15–22.
- Singh, S., Alawa, B., Mehra, S. S., Chakma, S., and Diwan, V. (2025). Microplastics in solid waste disposal sites of Central India: abundance and implications. *Sustain. Cities Soc.* 100002. doi: 10.1016/j.scsadv.2025.100002
- Srivastava, A. N., and Chakma, S. (2024). Quantification of landfill gas generation and energy recovery estimation from the municipal solid waste landfill sites of Delhi, India. *Energy Sourc. Part A Rec. Util. Environ. Eff.* 46, 7453–7466. doi: 10.1080/15567036.2020.1754970
- Swachh Survekshan Ranking. (2024). Available online at: <https://ss2023.sbmurban.org/#/ranking> (Accessed 12 June 2025).
- United Nations Environment Programme International Environment Technology Centre. (2022). Gender and waste management. Available online at: https://wedocs.unep.org/bitstream/handle/20.500.11822/41531/gender_waste_EN.pdf?sequence=4&isAllowed=y (Accessed 7 June 2025).
- Velijala, G. A. Sustainable organic waste management: a playbook for Lucknow, India. (2024). Available online at: <https://www.teriin.org/policy-brief/sustainable-organic-waste-management-playbook-lucknow-india> (Accessed 15 August 2025).
- Veliverronena, L., and Davidsone, A. (2025). The role of trust and perceived social justice in the waste recycling practices of households in Latvia. *Sustainability* 17:219. doi: 10.3390/su17010219
- Yuan, Q., and Zoungana, A. (2025). Systematic review of environmental and human health risk assessments in municipal solid waste management. *Disc. Sustain.* 6:930. doi: 10.1007/s43621-025-01544-8
- Yukalang, N., Clarke, B., and Ross, K. (2017). Barriers to effective municipal solid waste management in a rapidly urbanizing area in Thailand. *Int. J. Environ. Res. Public Health* 14:1013. doi: 10.3390/ijerph14091013
- Zhang, Z., Chen, Z., Zhang, J., Liu, Y., Chen, L., Yang, M., et al. (2024). Municipal solid waste management challenges in developing regions: a comprehensive review and future perspectives for Asia and Africa. *Sci. Total Environ.* 930:172794. doi: 10.1016/j.scitotenv.2024.172794